DATA ANALYTICS
MODULE END ASSIGNMENT 4_PYTHON -MARKETING CAMPAIGN PERFORMANCE INSIGHTS
DATE OF SUBMISSION:09.11.2026
NAME : DURGADEVI V

Marketing Campaign Performance Insights

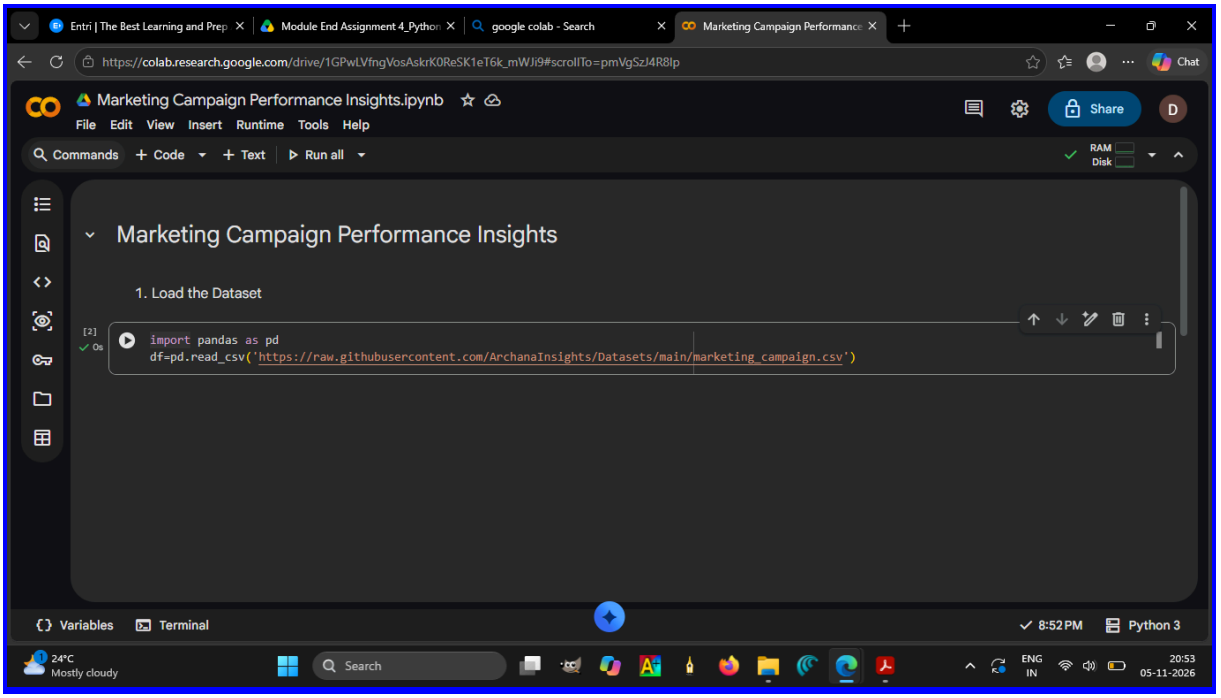
Problem Statement:

In the highly competitive landscape of digital marketing, effectively evaluating the success of various marketing campaigns is essential for optimizing return on investment (ROI) and improving overall performance. Despite having extensive data on multiple campaigns, there is a need for a thorough analysis to assess and compare key metrics such as conversion rates, acquisition costs, and ROI across different campaign types, channels, and audience segments. This project aims to uncover actionable insights by examining temporal trends, geographical influences, and audience responses to identify factors driving campaign success and provide recommendations for enhancing future marketing strategies.

Task 1:Load the Dataset

Solution :

Preview screenshot :



Explanation :

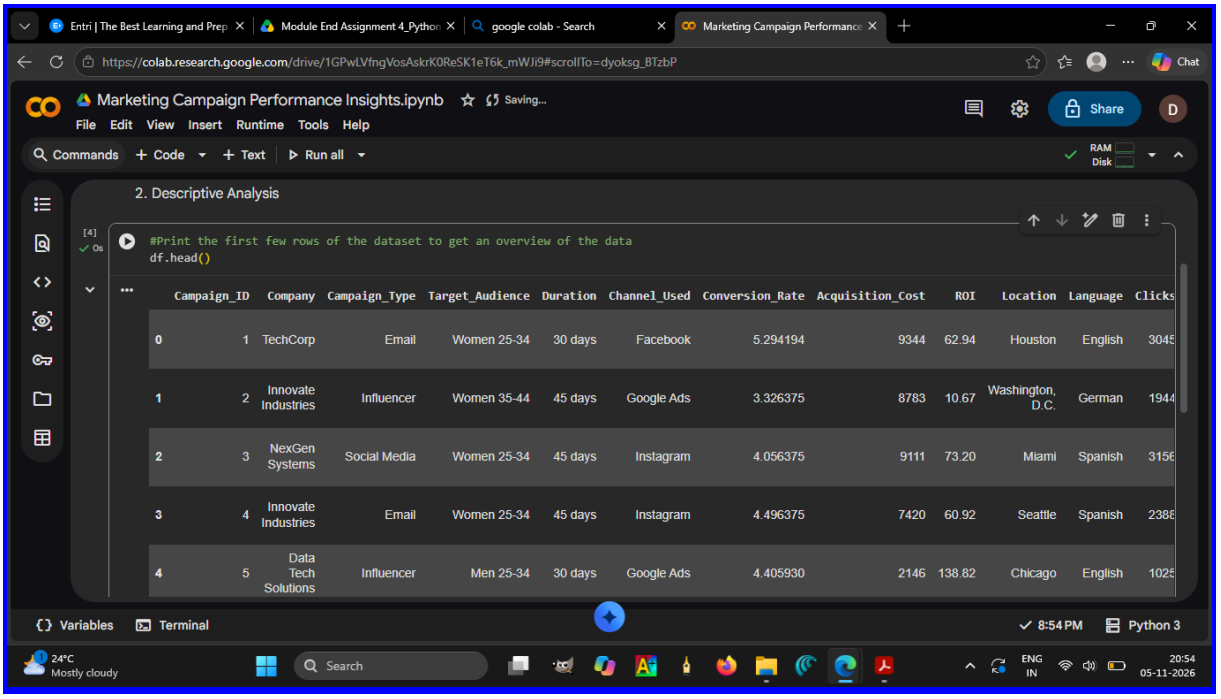
- First I had to load the data set by using pandas and I used the code ad df=pd.read_csv.
- This code will read the csv code and store that as a dataframe in df .
- The above mentioned screenshot represents the result.

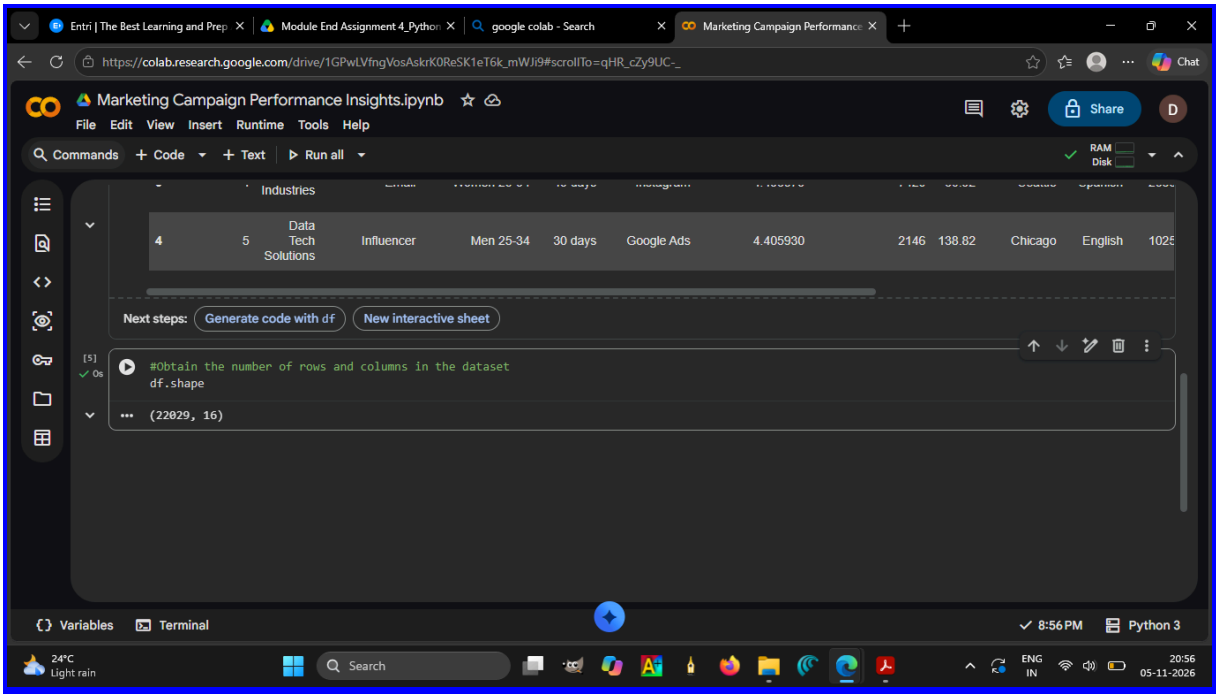
Task 2:Descriptive Analysis

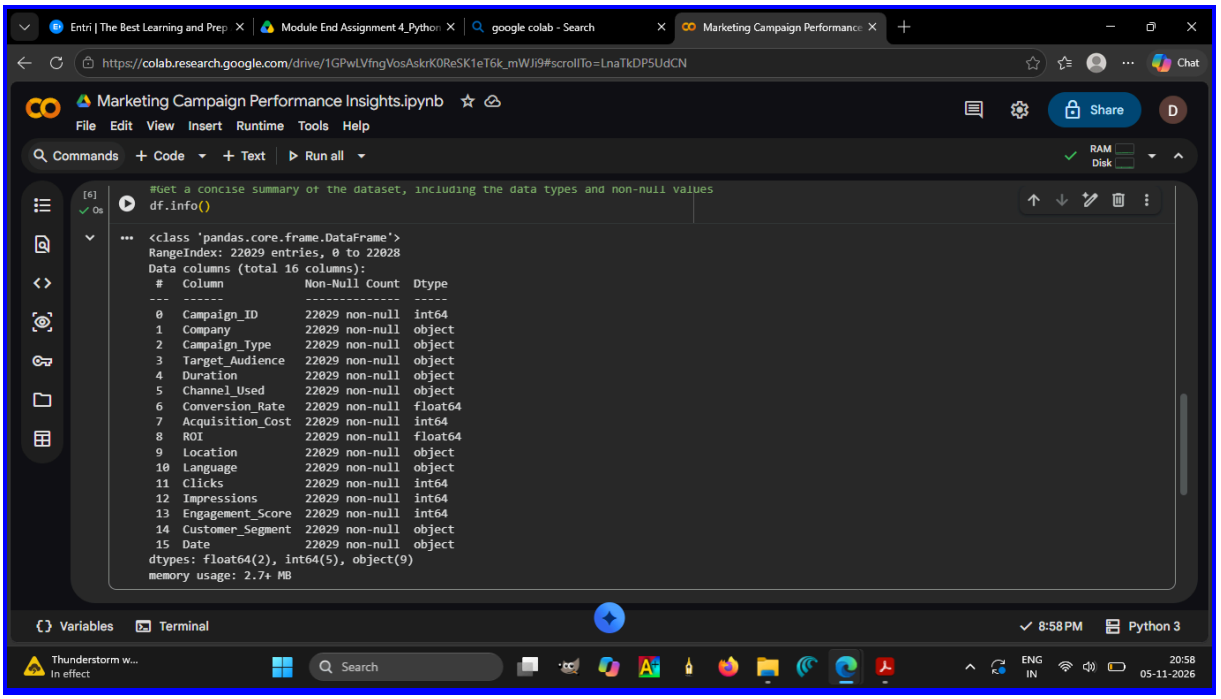
Solution :

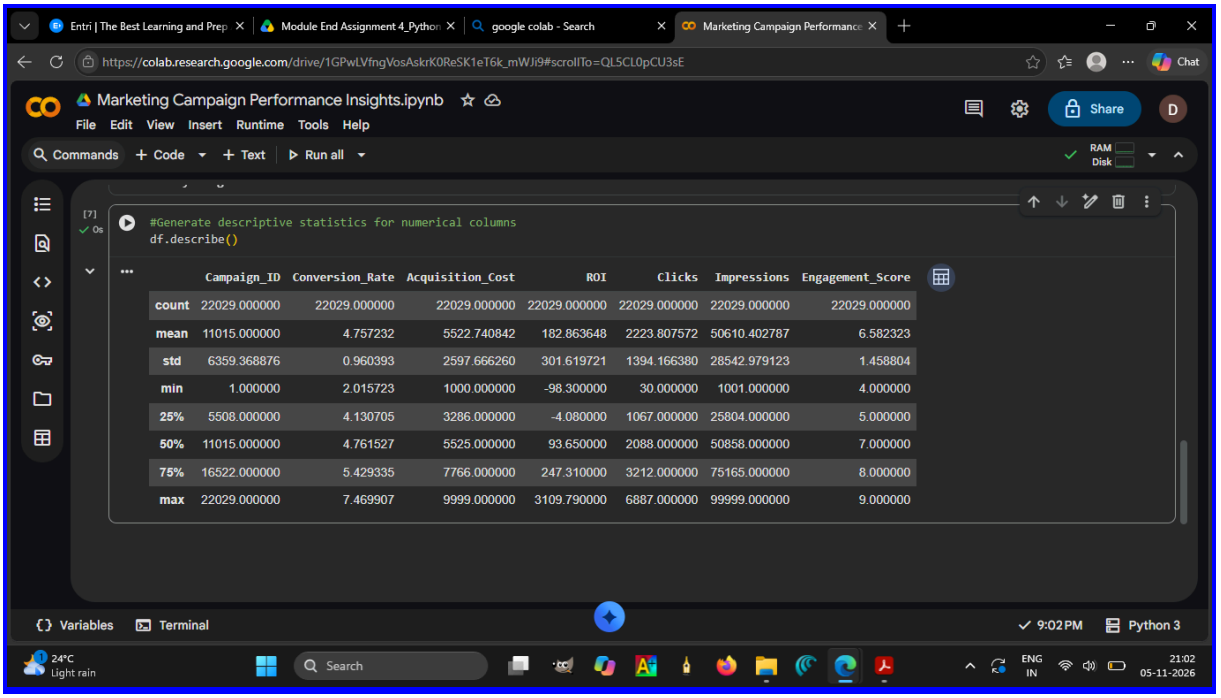
A)Basic Structure:

Preview screenshot :

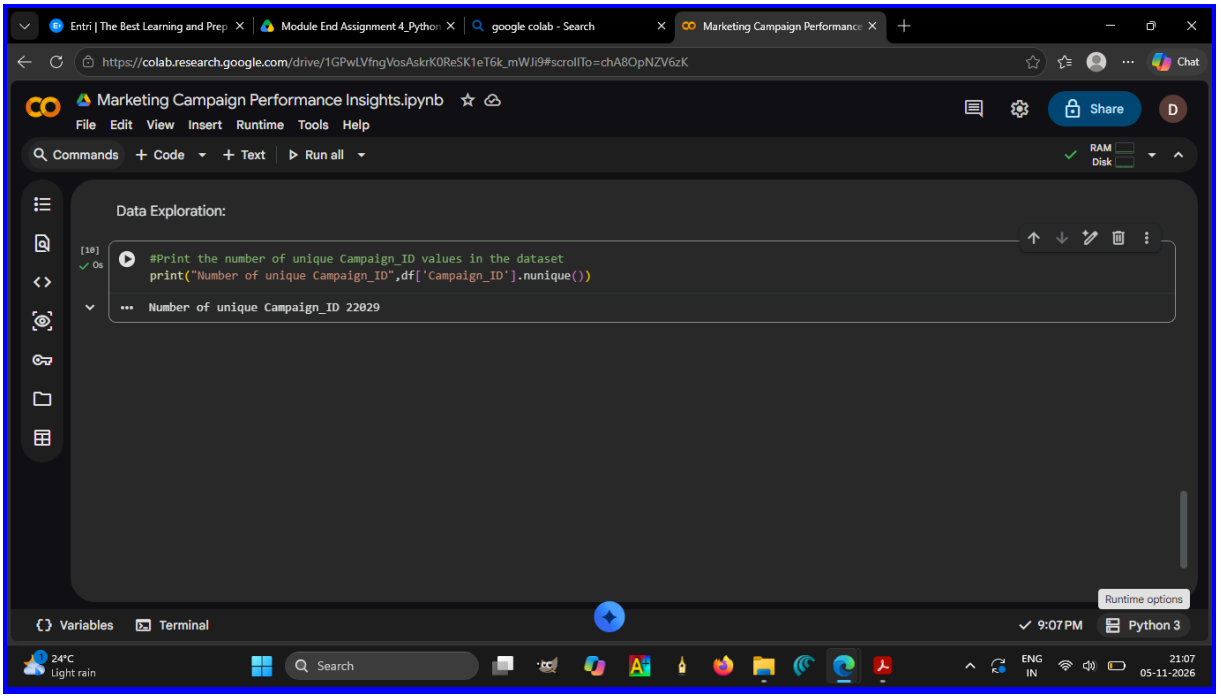


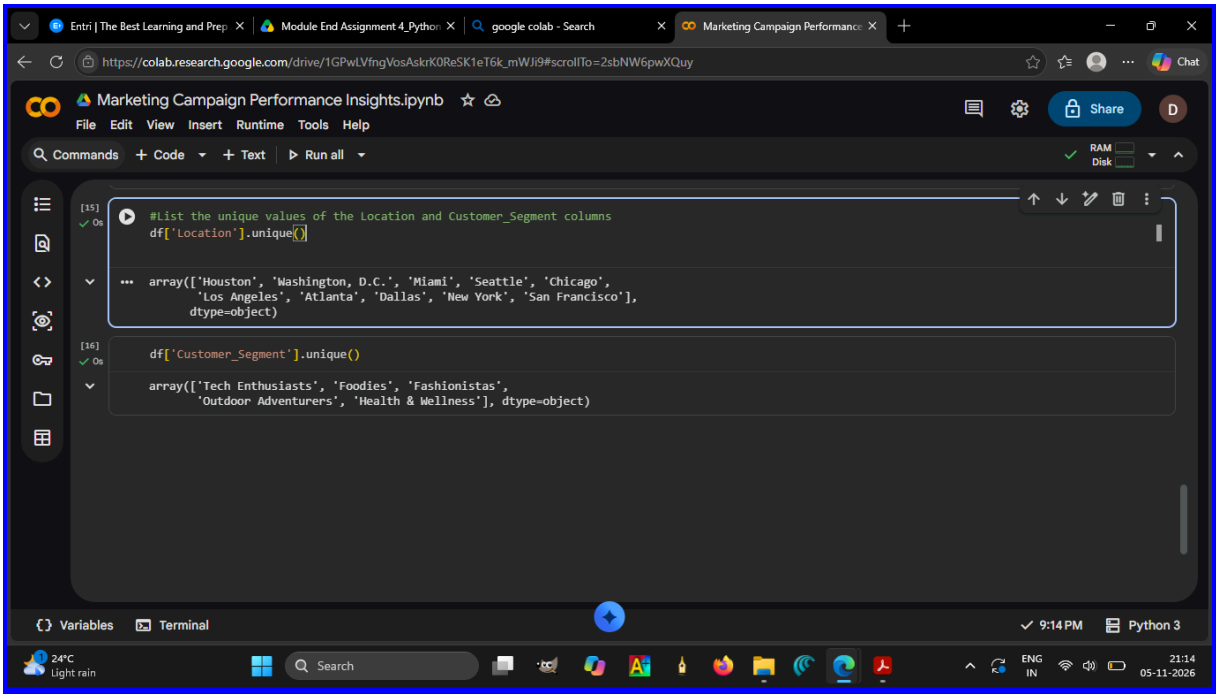

Marketing Campaign Performance Insights.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
RAM Disk
Industries
4 5 Data Tech Solutions Influencer Men 25-34 30 days Google Ads 4.405930 2146 138.82 Chicago English 1025
Next steps: Generate code with df New interactive sheet
[5] 0s
#Obtain the number of rows and columns in the dataset
df.shape
(22029, 16)
Variables Terminal
8:56 PM Python 3
24°C Light rain
Search
ENG IN
20:56 05-11-2026

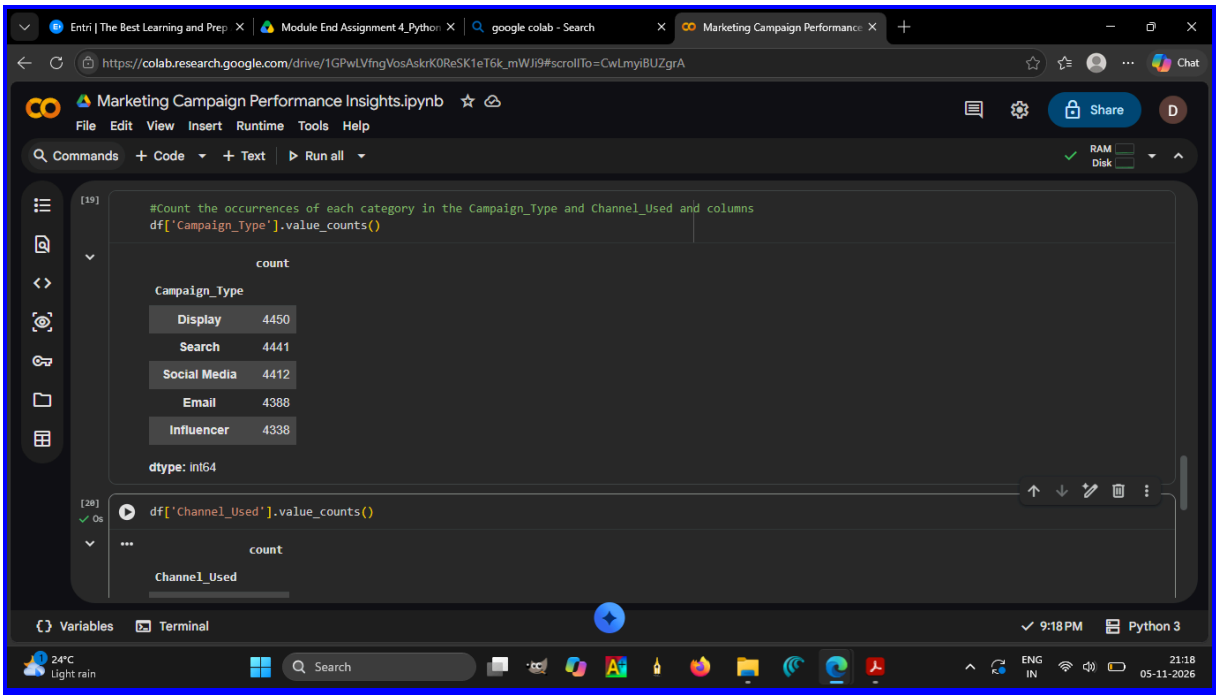

Marketing Campaign Performance Insights.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
RAM Disk
[6] 0s
#Get a concise summary of the dataset, including the data types and non-null values
df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 22029 entries, 0 to 22028
Data columns (total 16 columns):
Column Non-Null Count Dtype
0 Campaign_ID 22029 non-null int64
1 Company 22029 non-null object
2 Campaign_Type 22029 non-null object
3 Target_Audience 22029 non-null object
4 Duration 22029 non-null object
5 Channel_Used 22029 non-null object
6 Conversion_Rate 22029 non-null float64
7 Acquisition_Cost 22029 non-null int64
8 ROI 22029 non-null float64
9 Location 22029 non-null object
10 Language 22029 non-null object
11 Clicks 22029 non-null int64
12 Impressions 22029 non-null int64
13 Engagement_Score 22029 non-null int64
14 Customer_Segment 22029 non-null object
15 Date 22029 non-null object
dtypes: float64(2), int64(5), object(9)
memory usage: 2.7+ MB
Variables Terminal
8:58 PM Python 3
Thunderstorm w... In effect
Search
ENG IN
20:58 05-11-2026

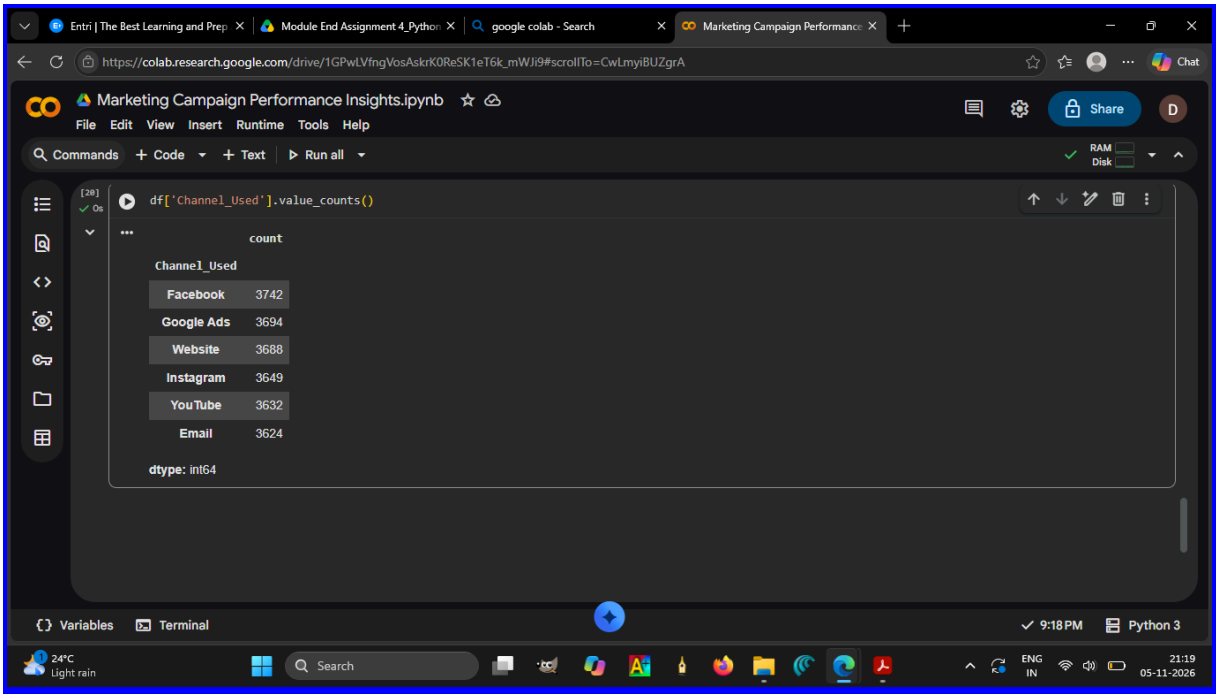


B)Data Exploration:




Marketing Campaign Performance Insights.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
#List the unique values of the Location and Customer_Segment columns
df['Location'].unique()
array(['Houston', 'Washington, D.C.', 'Miami', 'Seattle', 'Chicago',
'Los Angeles', 'Atlanta', 'Dallas', 'New York', 'San Francisco'],
dtype=object)
df['Customer_Segment'].unique()
array(['Tech Enthusiasts', 'Foodies', 'Fashionistas',
'Outdoor Adventurers', 'Health & Wellness'], dtype=object)
Variables Terminal
9:14 PM Python 3


Marketing Campaign Performance Insights.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
#Count the occurrences of each category in the Campaign_Type and Channel_Used and columns
df['Campaign_Type'].value_counts()
count
Campaign_Type
Display 4450
Search 4441
Social Media 4412
Email 4388
Influencer 4338
dtype: int64
df['Channel_Used'].value_counts()
count
Channel_Used
Variables Terminal
9:18 PM Python 3



Explanation :

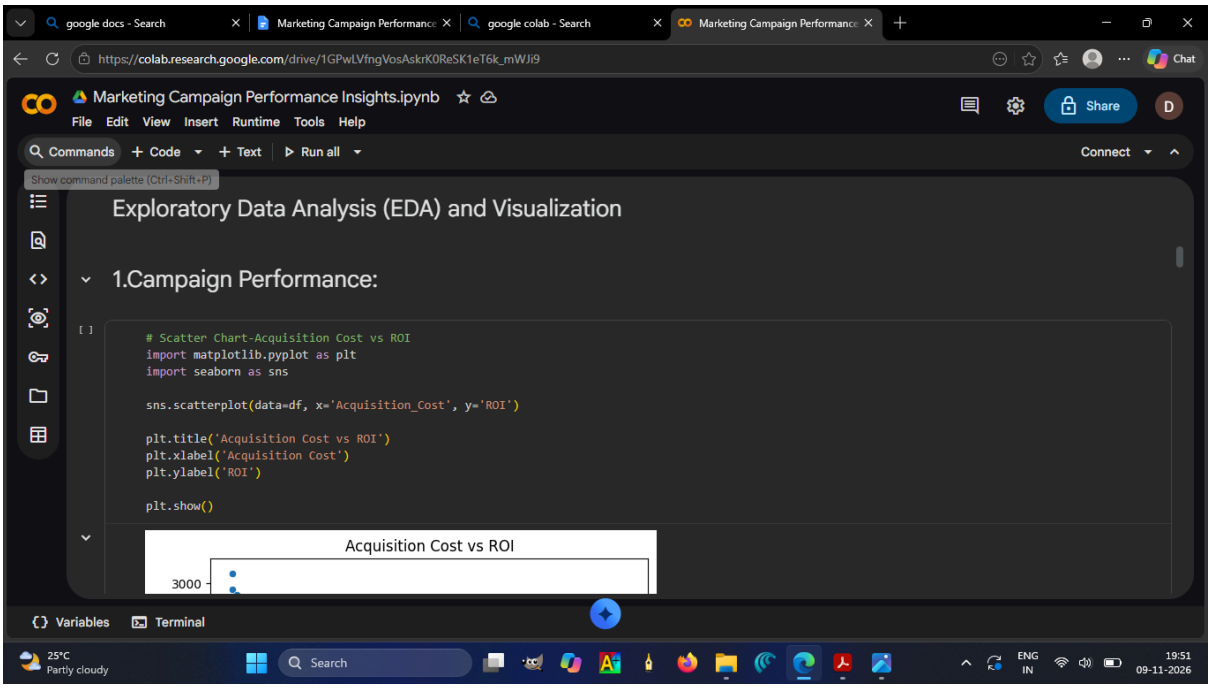
- First we need to print the first few rows of the dataset to get an overview of the data, so I used the head method to print the first 5 rows.
- To find the number of rows and columns in the dataset ,I used the shape method and that returned row count 22029 and column count is 16.
- To get a concise summary of the dataset, including the data types and non-null values , I used the info method. And that returns a perfect result.
- For generating descriptive statistics for numerical columns, I used the describe method. And that returned the results.
- For Data exploration , First I had printed the number of unique Campaign_ID values in the dataset by using the “nunique()“ method and get the result as number unique campaign id is 22029.
- To find the list of unique values of the Location and Customer_Segment columns,I used the unique() method .This returns a list of unique values .
- To count the occurrences of each category in the Campaign_Type and Channel_Used and columns. I used the value_counts() method to get the results.
- The above mentioned screenshots represent the result.

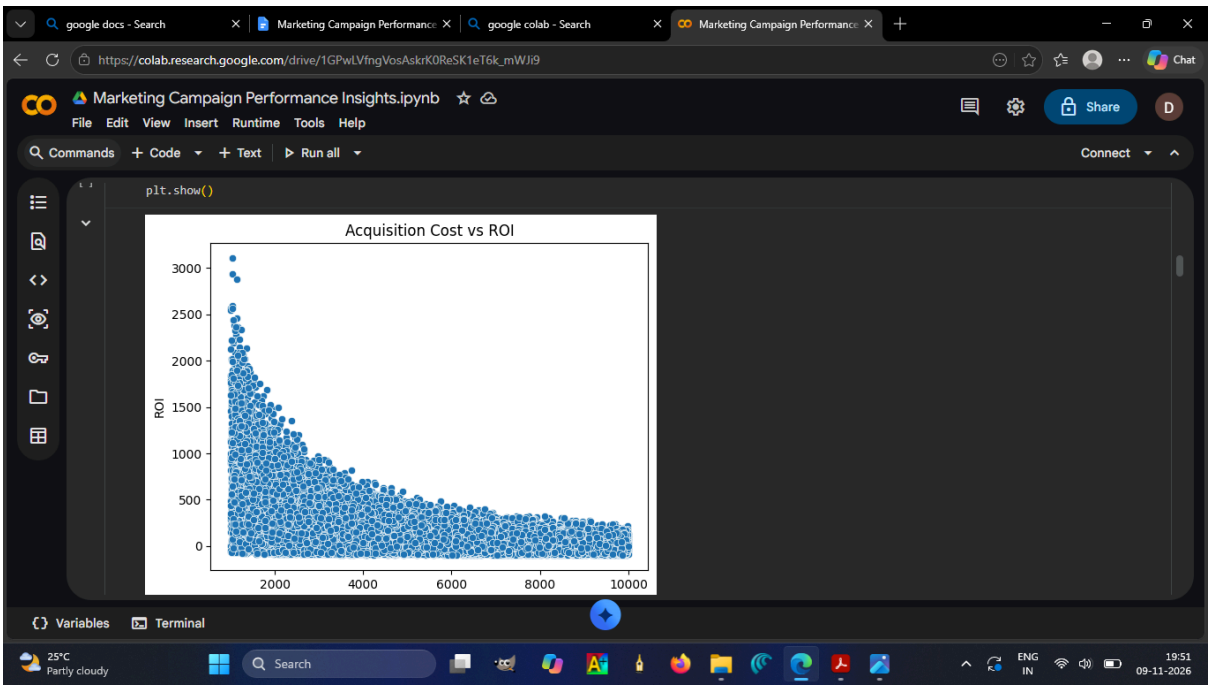
Exploratory Data Analysis (EDA) and Visualization

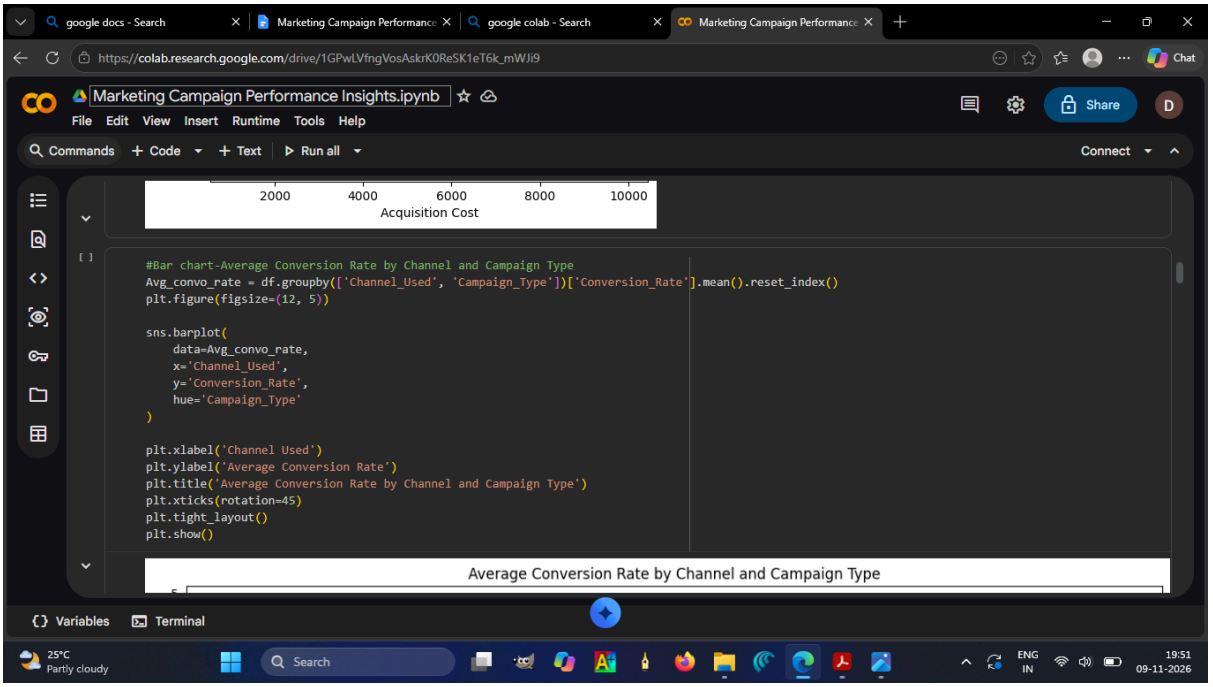
Task 1:Campaign Performance

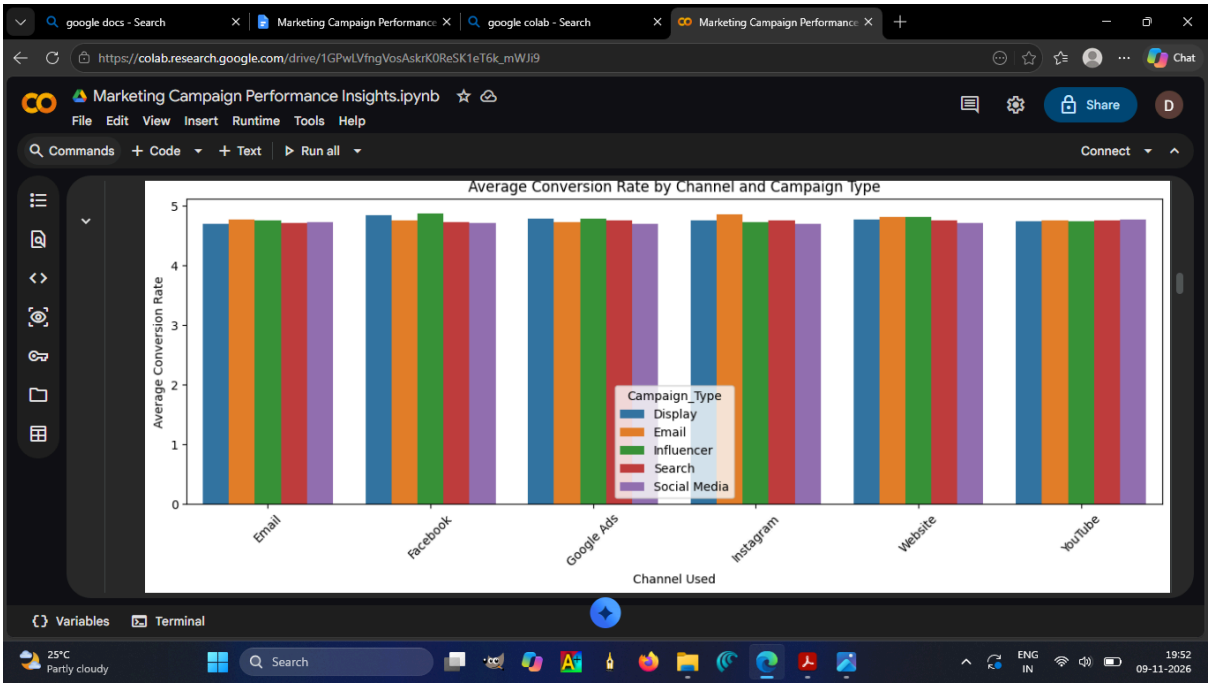
Solution :

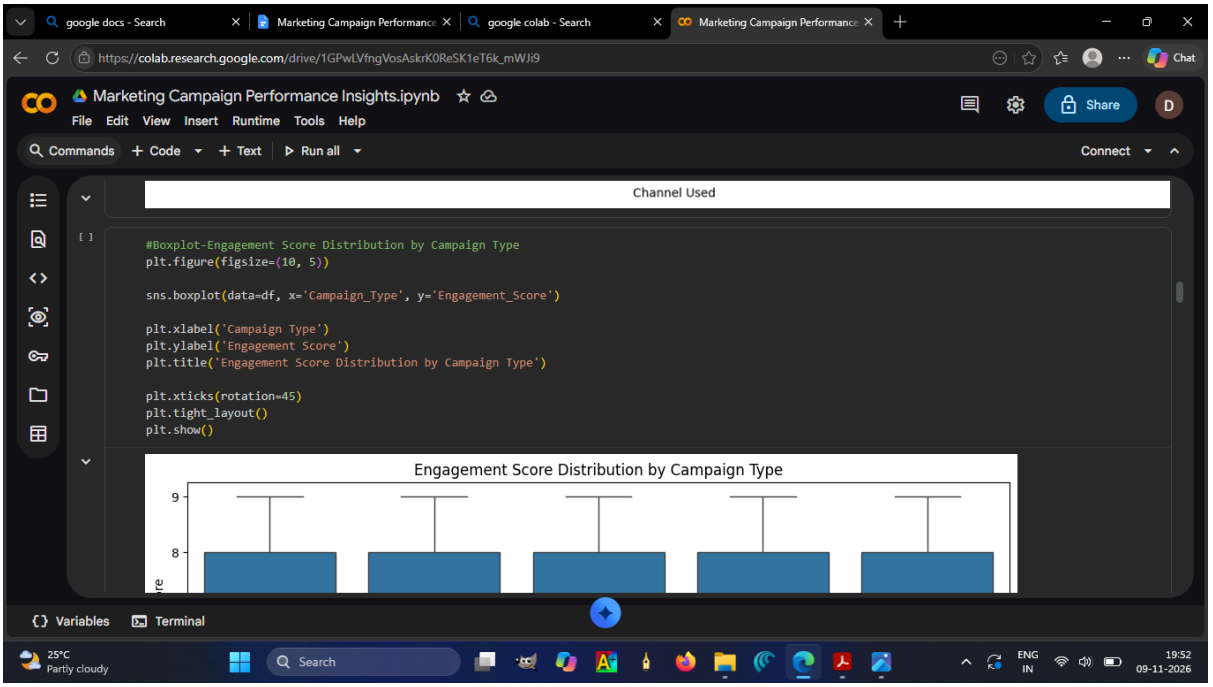
Preview screenshot :

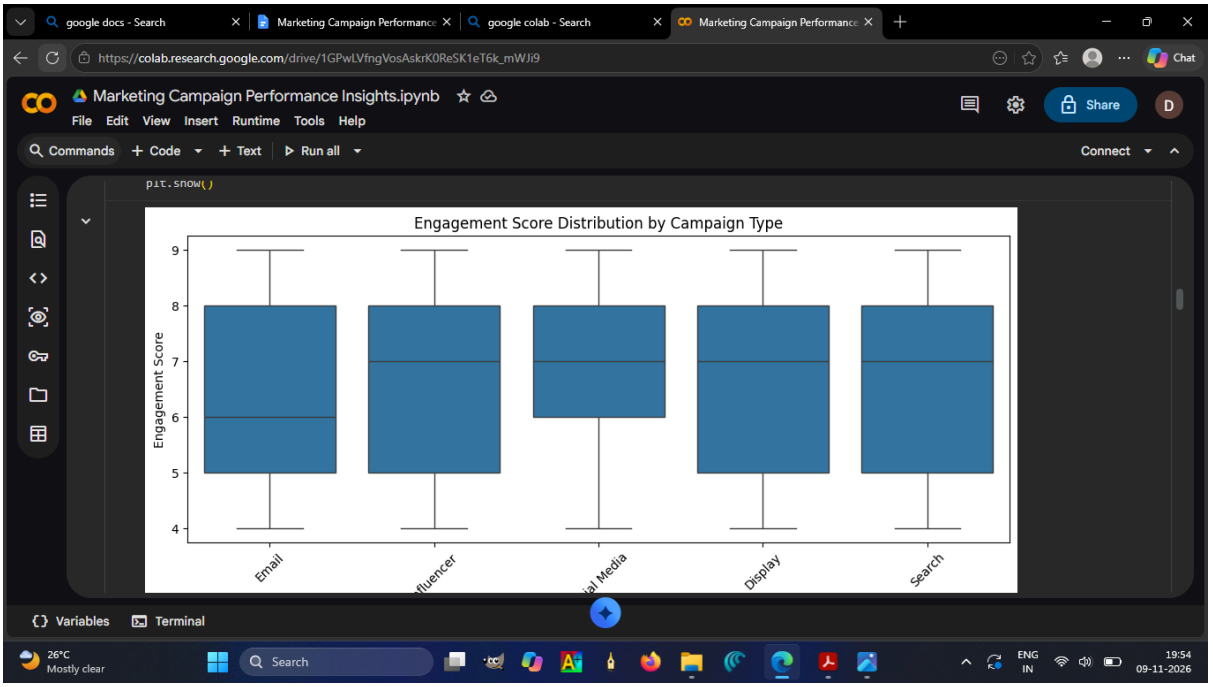


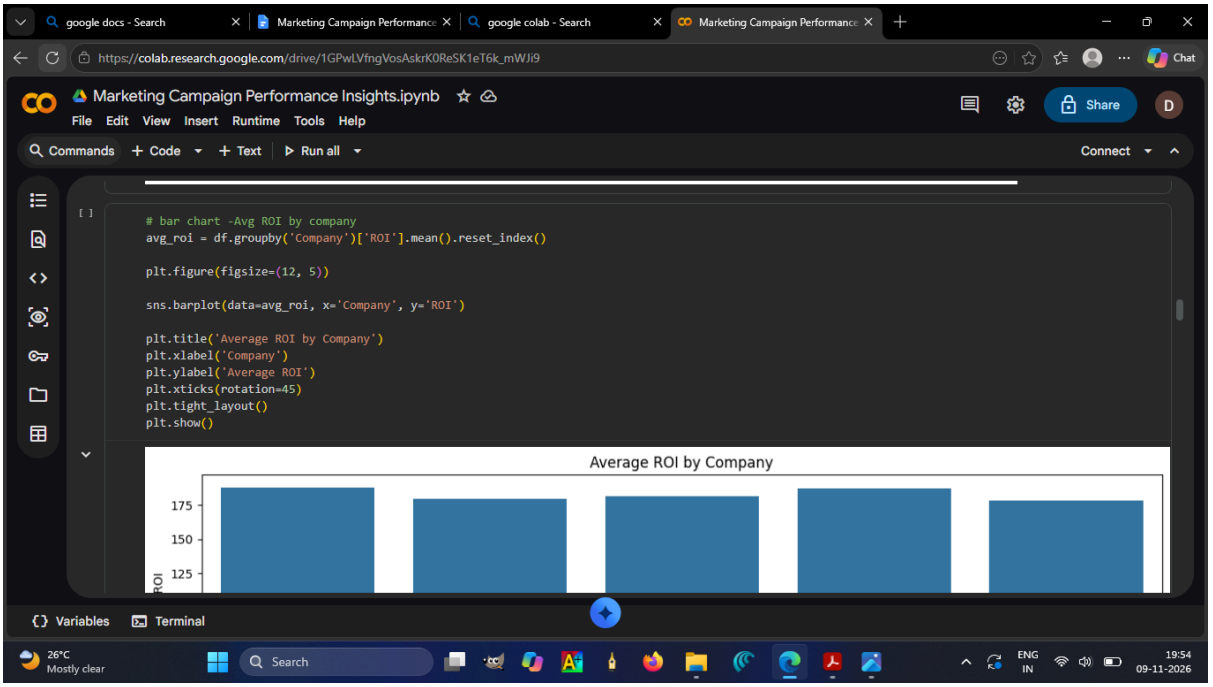


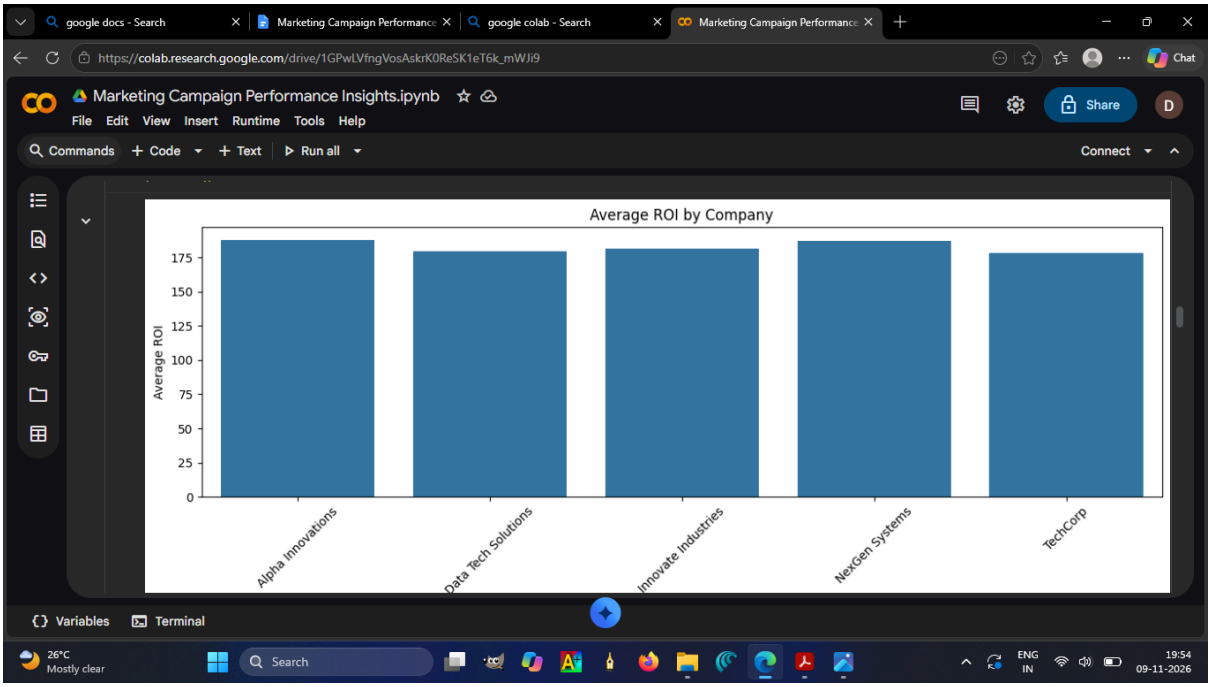

Marketing Campaign Performance Insights.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Connect
Share
2000 4000 6000 8000 10000
Acquisition Cost
#Bar chart-Average Conversion Rate by Channel and Campaign Type
Avg_convo_rate = df.groupby(['Channel_Used', 'Campaign_Type'])['Conversion_Rate'].mean().reset_index()
plt.figure(figsize=(12, 5))
sns.barplot(
data=Avg_convo_rate,
x='Channel_Used',
y='Conversion_Rate',
hue='Campaign_Type'
)
plt.xlabel('Channel Used')
plt.ylabel('Average Conversion Rate')
plt.title('Average Conversion Rate by Channel and Campaign Type')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
Average Conversion Rate by Channel and Campaign Type
Variables
Terminal

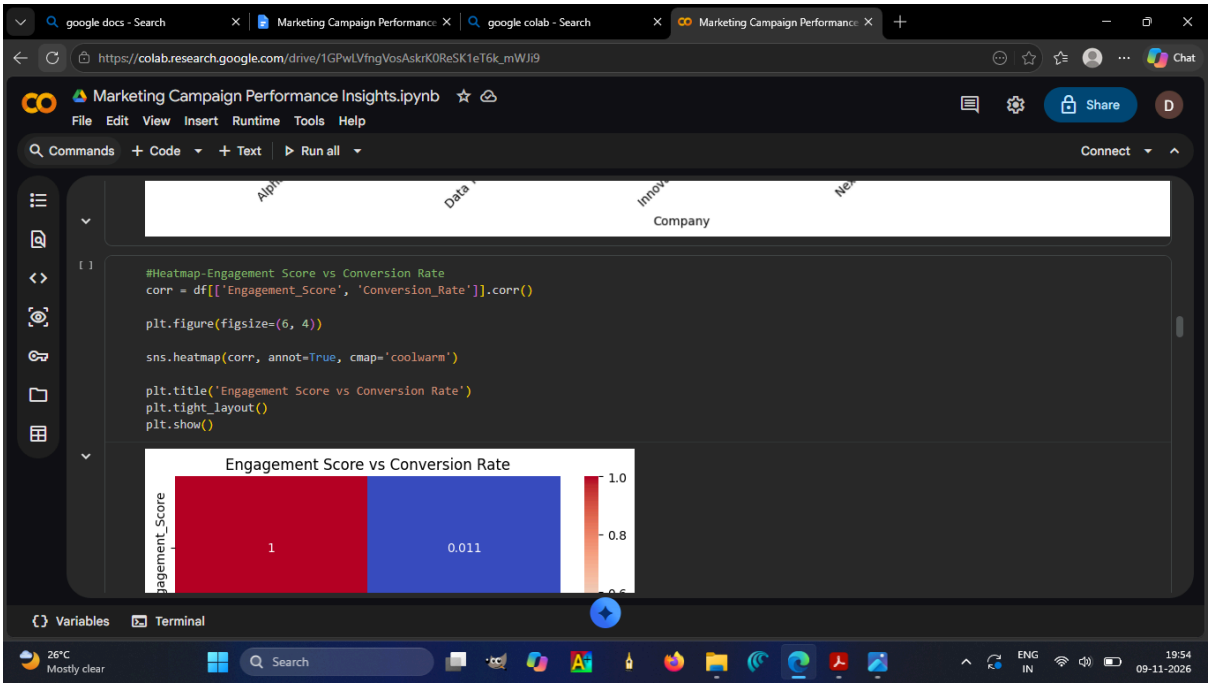

Marketing Campaign Performance Insights.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Connect
Share
Average Conversion Rate by Channel and Campaign Type
Average Conversion Rate
0
1
2
3
4
5
Campaign_Type
Display
Email
Influencer
Search
Social Media
Email
Facebook
Google Ads
Instagram
Website
YouTube
Channel Used
Variables
Terminal


Marketing Campaign Performance Insights.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Connect
Share
Channel Used
#Boxplot-Engagement Score Distribution by Campaign Type
plt.figure(figsize=(10, 5))
sns.boxplot(data=df, x='Campaign_Type', y='Engagement_Score')
plt.xlabel('Campaign Type')
plt.ylabel('Engagement Score')
plt.title('Engagement Score Distribution by Campaign Type')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
Engagement Score Distribution by Campaign Type
Variables
Terminal


Marketing Campaign Performance Insights.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Connect
Share
plt.show()
Engagement Score Distribution by Campaign Type
Engagement Score
Email
Display
Search
Variables
Terminal


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Marketing Campaign Performance
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Marketing Campaign Performance
https://colab.research.google.com/drive/1GPwLVfngVosAskrK0ReSK1eT6k_mWJi9
Chat
Marketing Campaign Performance Insights.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
Connect
[ ]
bar chart -Avg ROI by company
avg_roi = df.groupby('Company')['ROI'].mean().reset_index()
plt.figure(figsize=(12, 5))
sns.barplot(data=avg_roi, x='Company', y='ROI')
plt.title('Average ROI by Company')
plt.xlabel('Company')
plt.ylabel('Average ROI')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
Average ROI by Company
175
150
125
ROI
Variables
Terminal
26°C
Mostly clear
Search
ENG
IN
19:54
09-11-2026


google docs - Search
Marketing Campaign Performance
google colab - Search
Marketing Campaign Performance
https://colab.research.google.com/drive/1GPwLVfngVosAskrK0ReSK1eT6k_mWJi9
Chat
Marketing Campaign Performance Insights.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
Connect
Average ROI by Company
175
150
125
100
75
50
25
0
Average ROI
Alpha Innovations
Data Tech Solutions
Innovate Industries
NexGen Systems
TechCorp
Variables
Terminal
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Explanation :

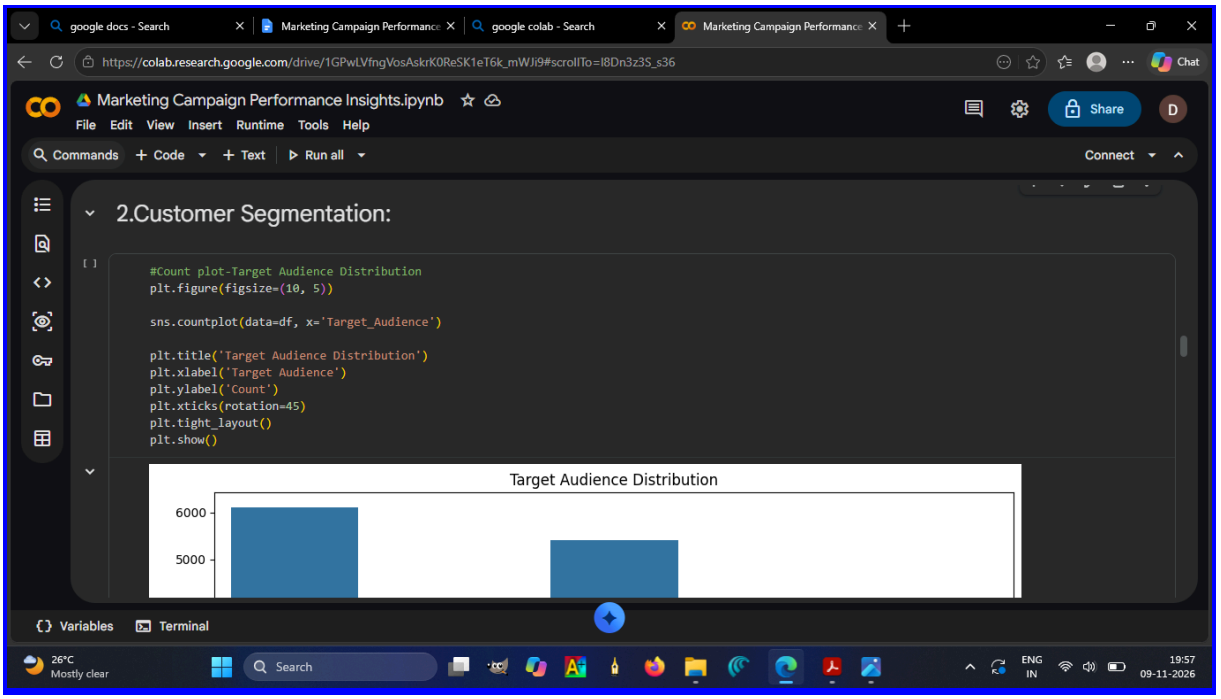
- For creating a scatter plot to visualize the relationship between Acquisition_Cost and ROI ,I had imported seaborn and matplotlib and then used acquisition cost in x axis and ROI in y axis It will find the relationship between acquisition cost and ROI then get a scatter plot as result.
- For creating a bar chart to visualize the average Conversion_Rate for different Channel_Used, categorized by Campaign_Type. I used a mean method for finding average conversion rate, then created a bar chart by assigning channels used in x axis ,conversion rate in y axis with campaign type in hue and got the result .

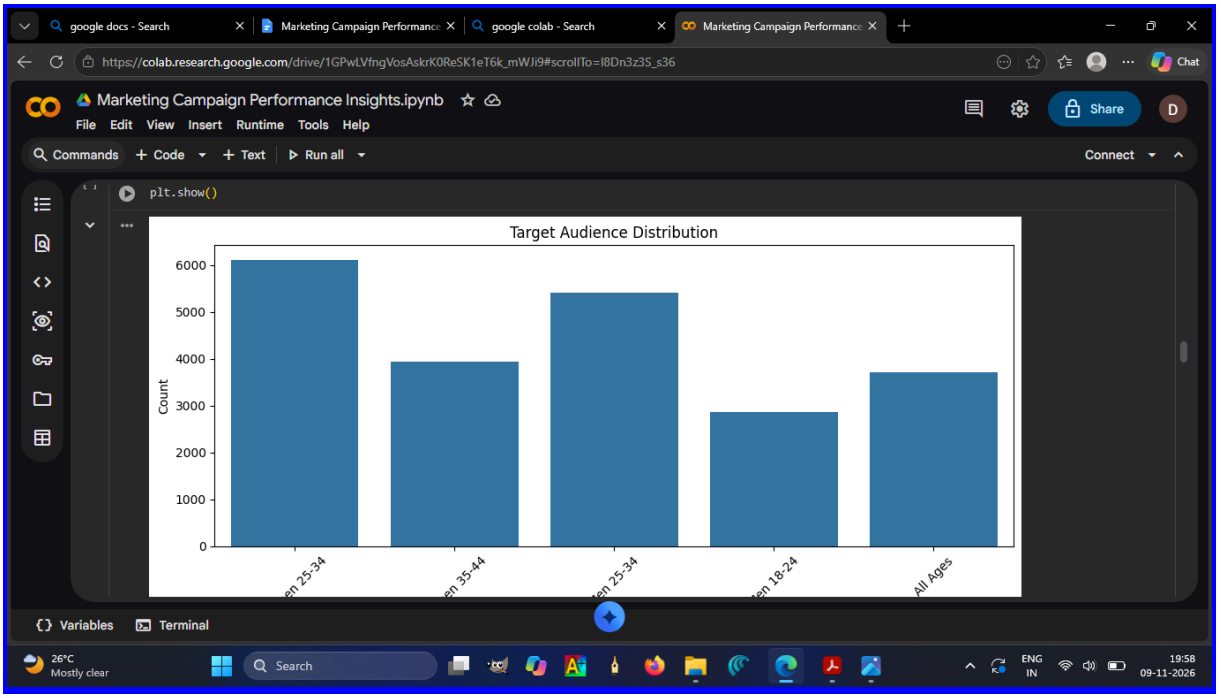
- For visualizing the distribution of Engagement_Score across different Campaign_Type by using a box plot .I used campaign type in x axis and engagement score in y axis then got a result of box plot.
- For analyzing the average ROI by Company using a bar chart to compare the profitability of campaigns conducted by different companies. First I had find average ROI by using mean() method and assign company in x axis and ROI in y axis and get the bar chart as result
- To examine the correlation between Engagement_Score and Conversion_Rate by using a heatmap, First I calculated the correlation by using corr() method and created a headmap.
- The above mentioned screenshots represent the results.

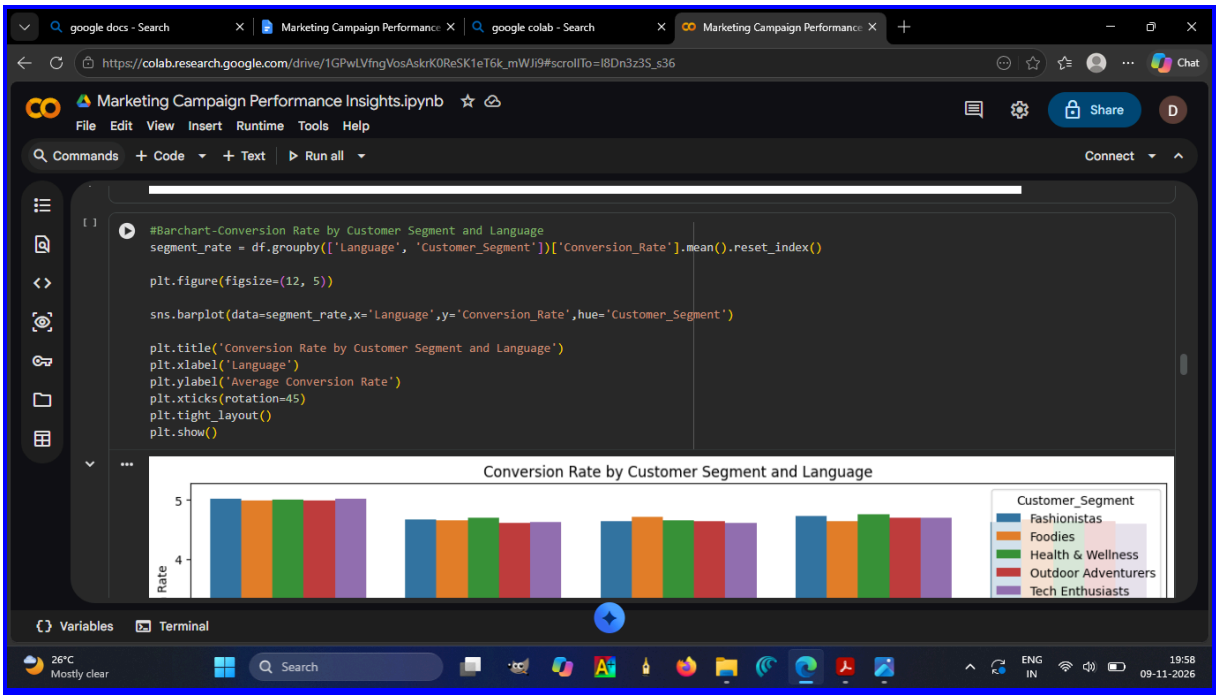
Task 2 : Customer Segmentation

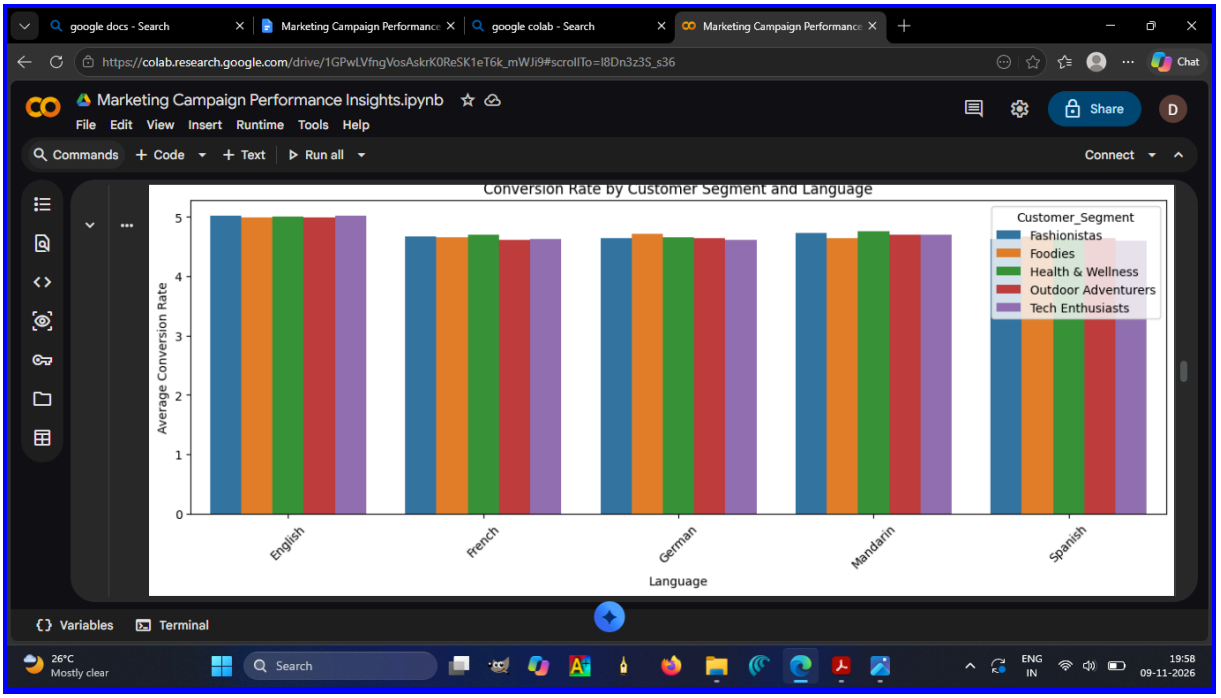
Solution :

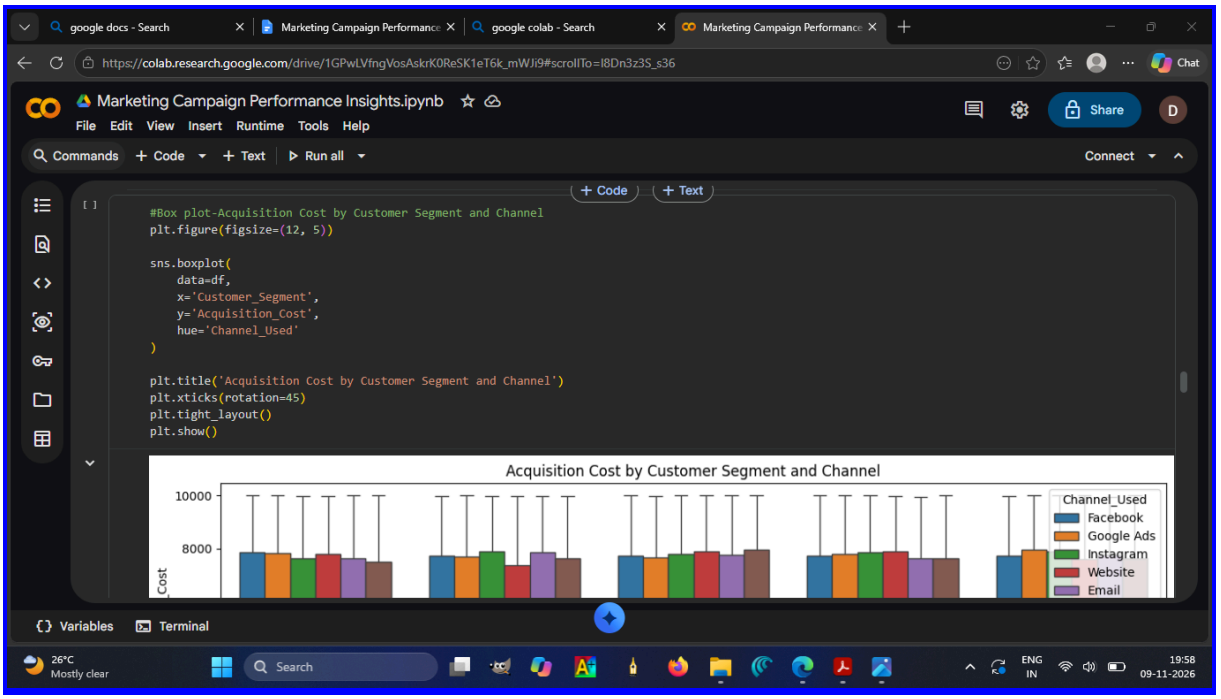
Preview screenshot :

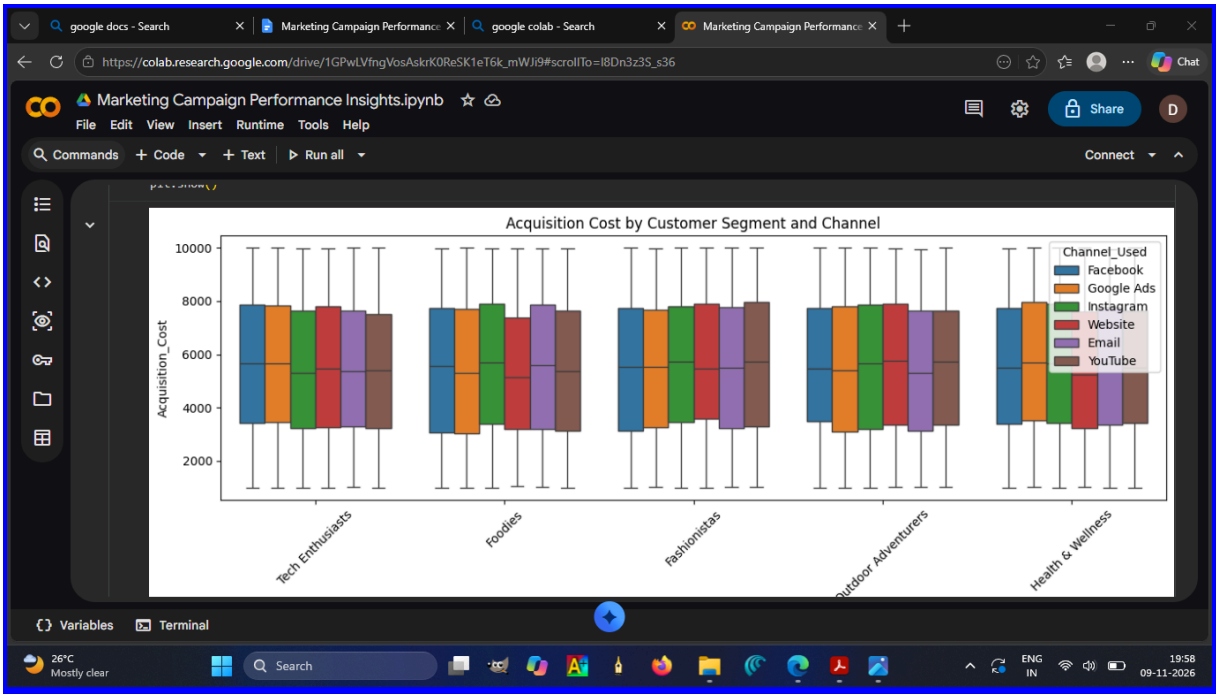


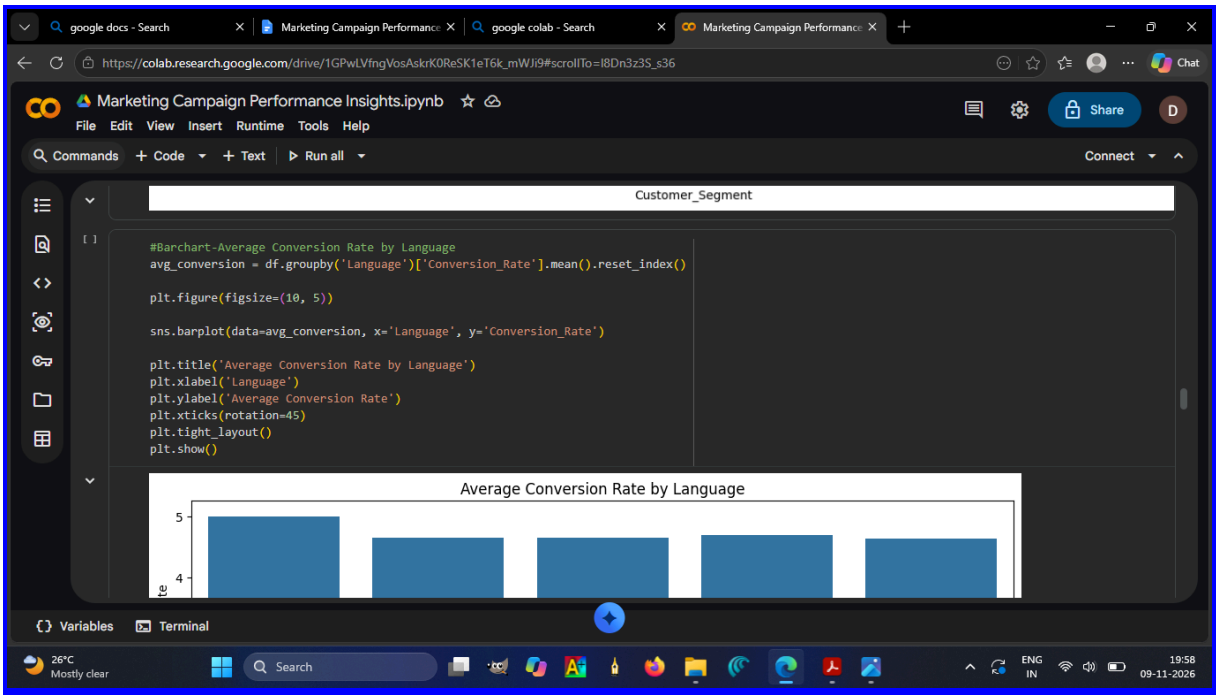

Marketing Campaign Performance Insights.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
Connect
plt.show()
Target Audience Distribution
Count
6000
5000
4000
3000
2000
1000
0
en 25-34
en 35-44
en 25-34
en 18-24
All Ages
Variables
Terminal
26°C
Mostly clear
Search
ENG
IN
19:58
09-11-2026

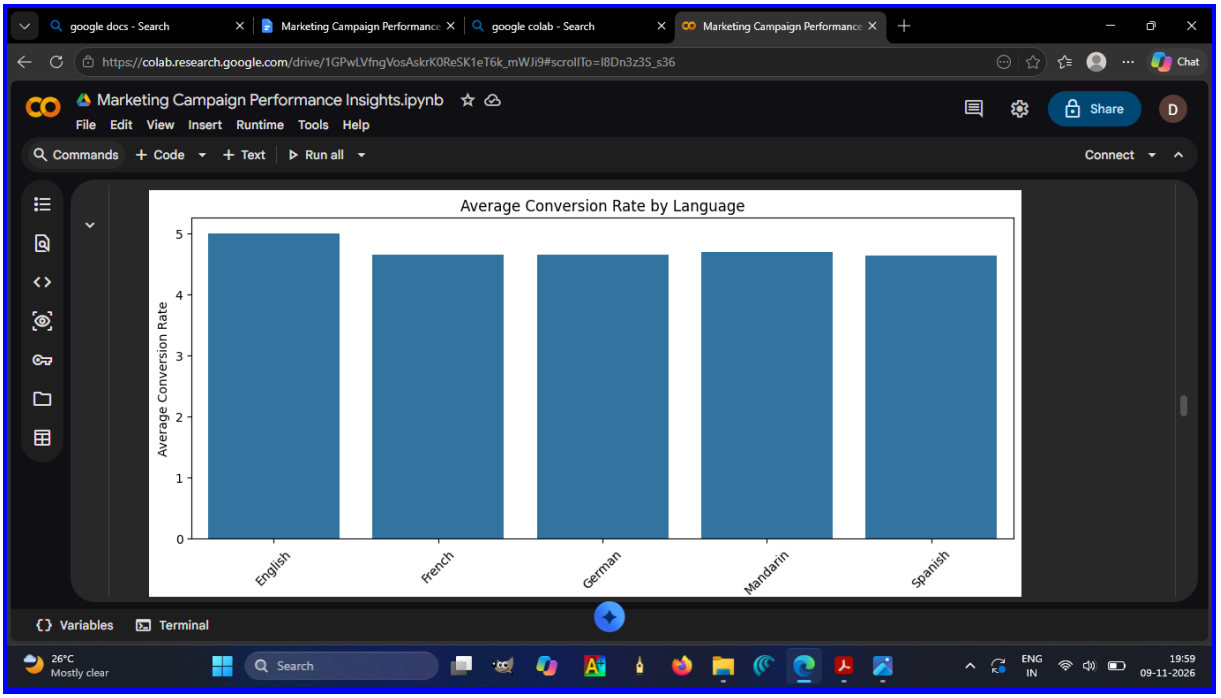

Marketing Campaign Performance Insights.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
Connect
#Barchart-Conversion Rate by Customer Segment and Language
segment_rate = df.groupby(['Language', 'Customer_Segment'])['Conversion_Rate'].mean().reset_index()
plt.figure(figsize=(12, 5))
sns.barplot(data=segment_rate,x='Language',y='Conversion_Rate',hue='Customer_Segment')
plt.title('Conversion Rate by Customer Segment and Language')
plt.xlabel('Language')
plt.ylabel('Average Conversion Rate')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
Conversion Rate by Customer Segment and Language
Rate
5
4
Customer_Segment
Fashionistas
Foodies
Health & Wellness
Outdoor Adventurers
Tech Enthusiasts
Variables
Terminal
26°C
Mostly clear
Search
ENG
IN
19:58
09-11-2026


Marketing Campaign Performance Insights.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
Connect
Conversion Rate by Customer Segment and Language
Average Conversion Rate
Language
English
French
German
Mandarin
Spanish
Customer_Segment
Fashionistas
Foodies
Health & Wellness
Outdoor Adventurers
Tech Enthusiasts
Variables
Terminal


Marketing Campaign Performance Insights.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
Connect
#Box plot-Acquisition Cost by Customer Segment and Channel
plt.figure(figsize=(12, 5))
sns.boxplot(
data=df,
x='Customer_Segment',
y='Acquisition_Cost',
hue='Channel_Used'
)
plt.title('Acquisition Cost by Customer Segment and Channel')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
Acquisition Cost by Customer Segment and Channel
Cost
Channel_Used
Facebook
Google Ads
Instagram
Website
Email
Variables
Terminal


Marketing Campaign Performance Insights.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
Connect
Acquisition Cost by Customer Segment and Channel
Acquisition_Cost
10000
8000
6000
4000
2000
Channel_Used
Facebook
Google Ads
Instagram
Website
Email
YouTube
Tech Enthusiasts
Foodies
Fashionistas
Health & Wellness
Variables
Terminal
26°C
Mostly clear
Search
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IN
19:58
09-11-2026


Marketing Campaign Performance Insights.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
Connect
Customer_Segment
#Barchart-Average Conversion Rate by Language
avg_conversion = df.groupby('Language')['Conversion_Rate'].mean().reset_index()
plt.figure(figsize=(10, 5))
sns.barplot(data=avg_conversion, x='Language', y='Conversion_Rate')
plt.title('Average Conversion Rate by Language')
plt.xlabel('Language')
plt.ylabel('Average Conversion Rate')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
Average Conversion Rate by Language
5
4
Variables
Terminal
26°C
Mostly clear
Search
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09-11-2026



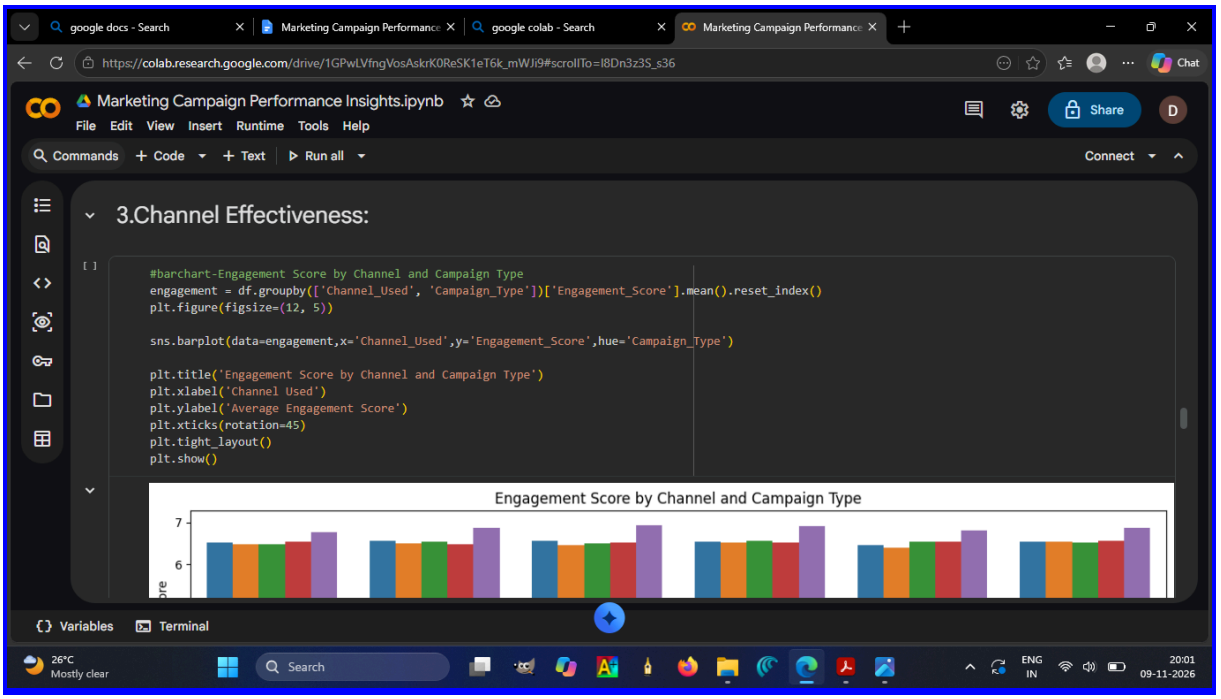
Explanation:

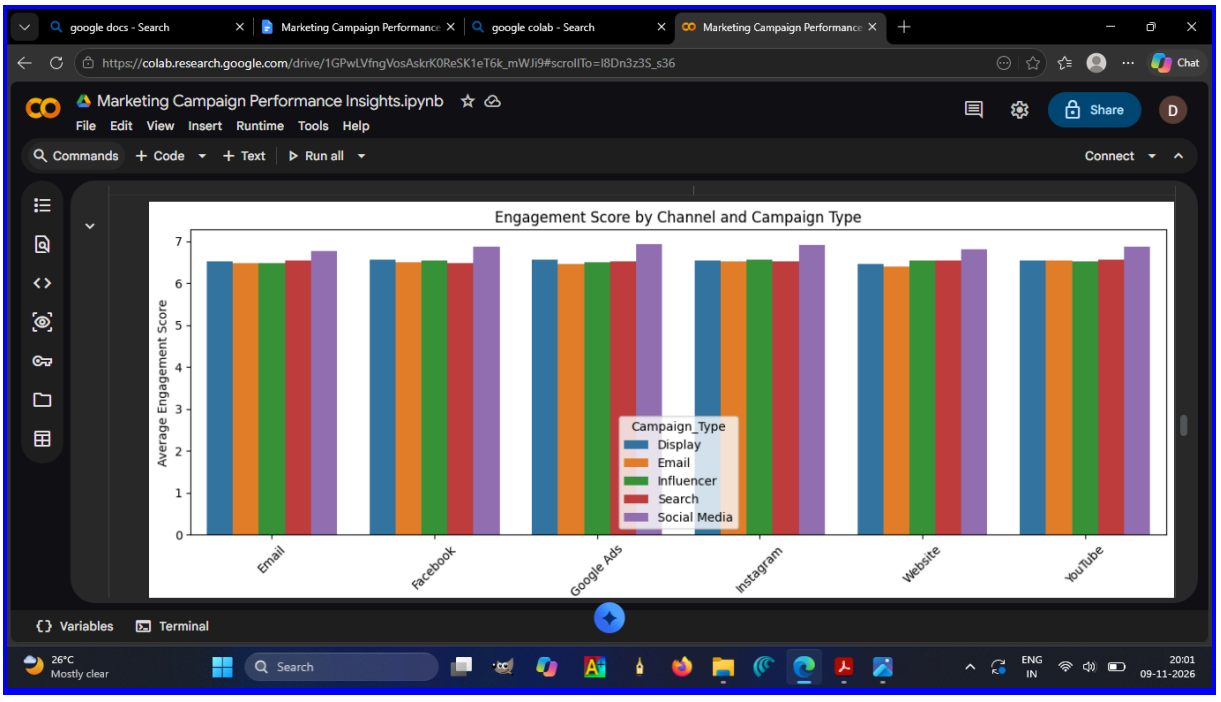
- For creating a count plot to visualize the distribution of Target_Audience . I used the target audience in x axis and get the result.
- Identify which Customer_Segment has the highest Conversion_Rate for each Language by using a bar chart and calculating segment rate by using mean and groupby method and get the result .
- To visualize the distribution of Acquisition_Cost across each Customer_Segment, categorized by Channel_Used, using a box plot by assigning customer segment in x axis and acquisition cost in y axis and channel used in hue.
- For analyze average Conversion_Rate by Language using a bar chart to compare the effectiveness of campaigns conducted in different languages by assigning language in x axis and conversion rate in y axis.
- The above mentioned screenshots represent the results.

Task 3: Channel Effectiveness

Solution :

Preview screenshot :





google docs - Search | Marketing Campaign Performance | google colab - Search | Marketing Campaign Performance

https://colab.research.google.com/drive/1GPwLVfngVosAskrK0ReSK1eT6k_mWJi9#scrollTo=l8Dn3z3S_s36

Marketing Campaign Performance Insights.ipynb

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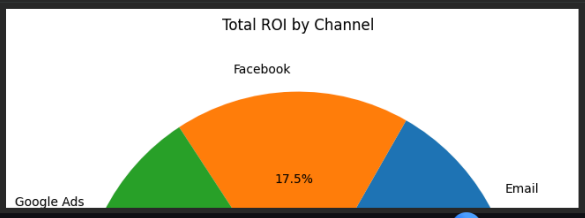
Channel Used

```
#Pie chart-Total ROI by Channel
channel_roi = df.groupby('Channel_Used')['ROI'].sum()

plt.figure(figsize=(8, 8))

plt.pie(channel_roi,labels=channel_roi.index,autopct='%1.1f%%')

plt.title('Total ROI by Channel')
plt.show()
```



Variables Terminal

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google docs - Search | Marketing Campaign Performance | google colab - Search | Marketing Campaign Performance

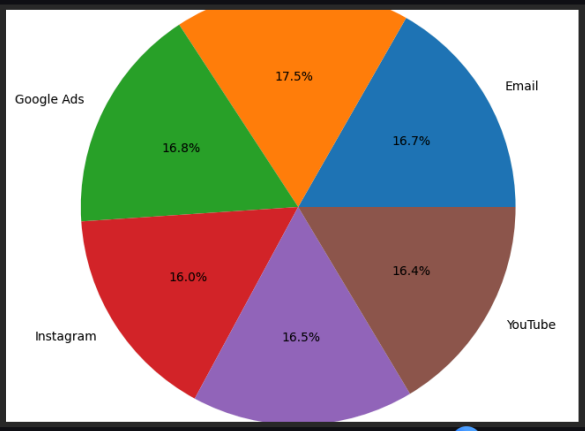
https://colab.research.google.com/drive/1GPwLVfngVosAskrK0ReSK1eT6k_mWJi9#scrollTo=l8Dn3z3S_s36

Marketing Campaign Performance Insights.ipynb

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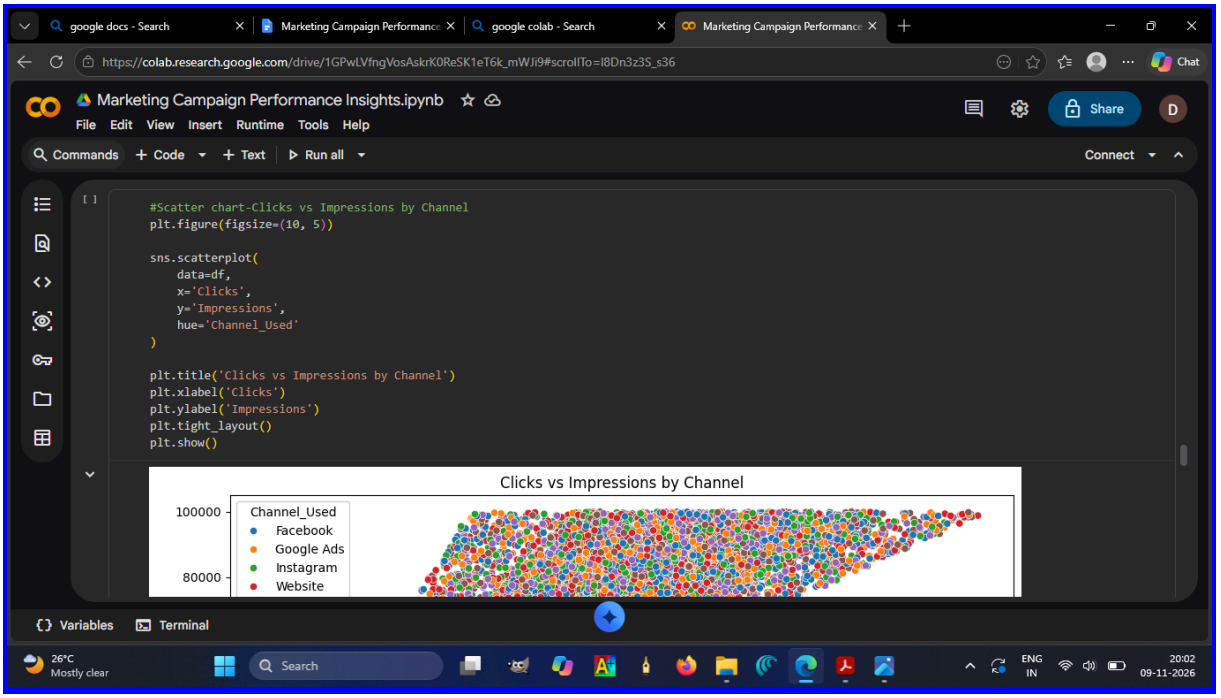
Commands + Code + Text Run all Connect

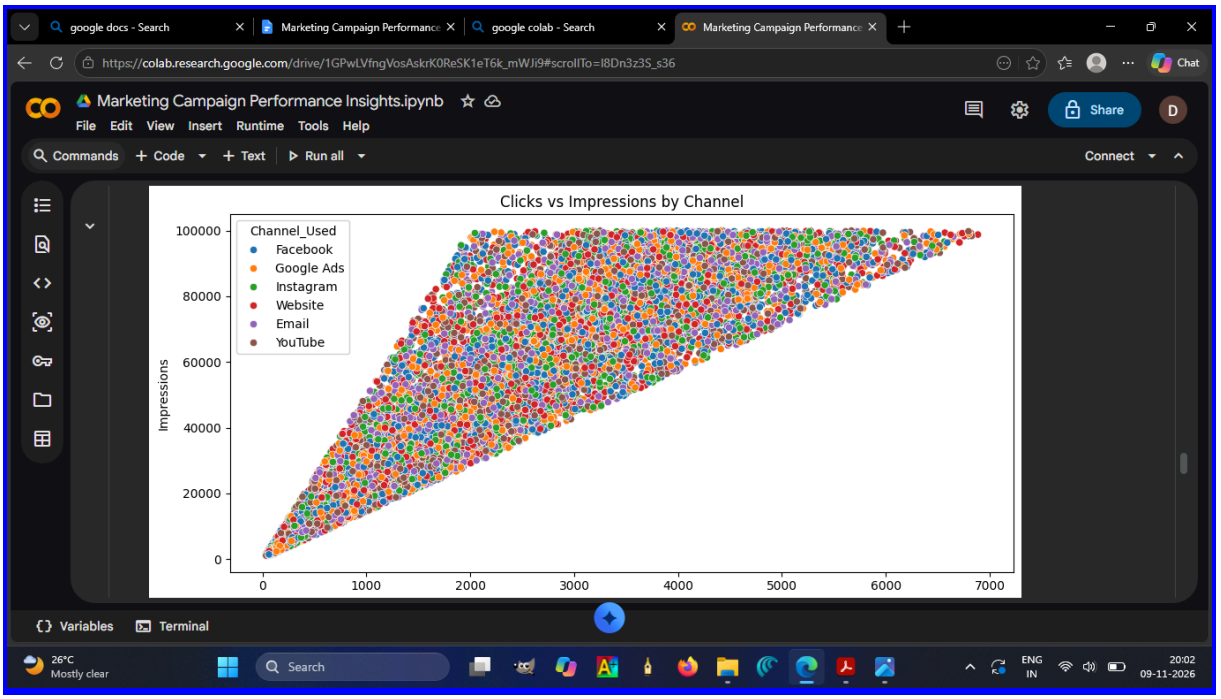
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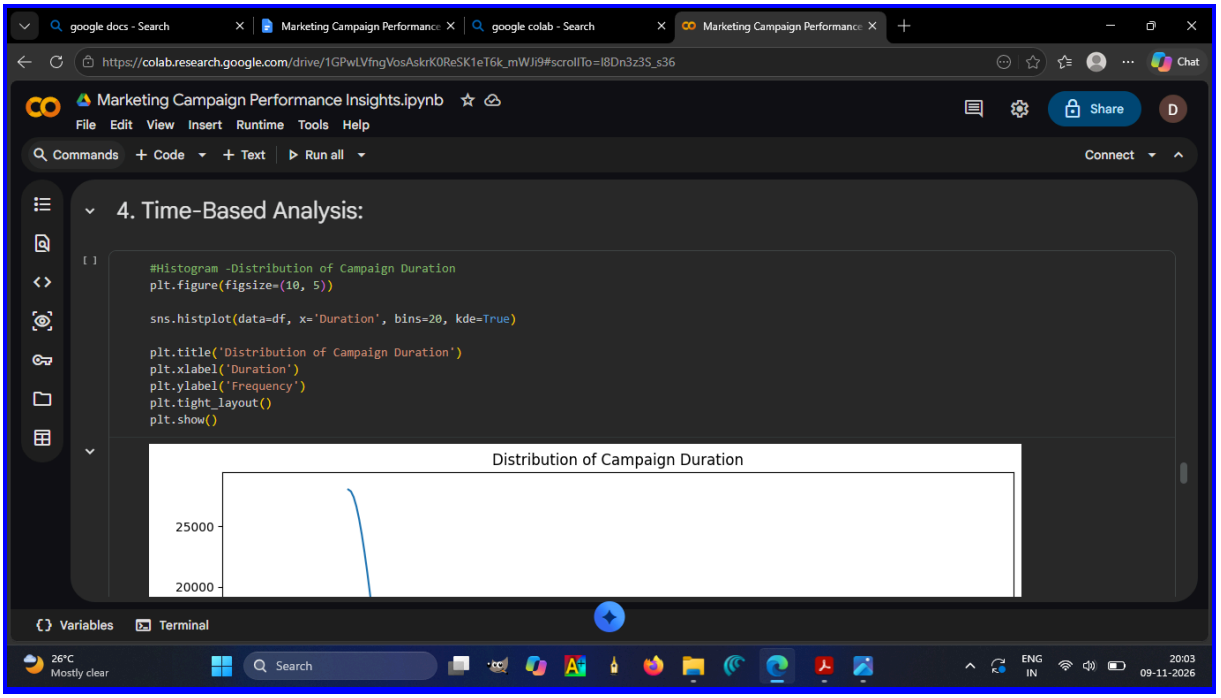
Explanation :

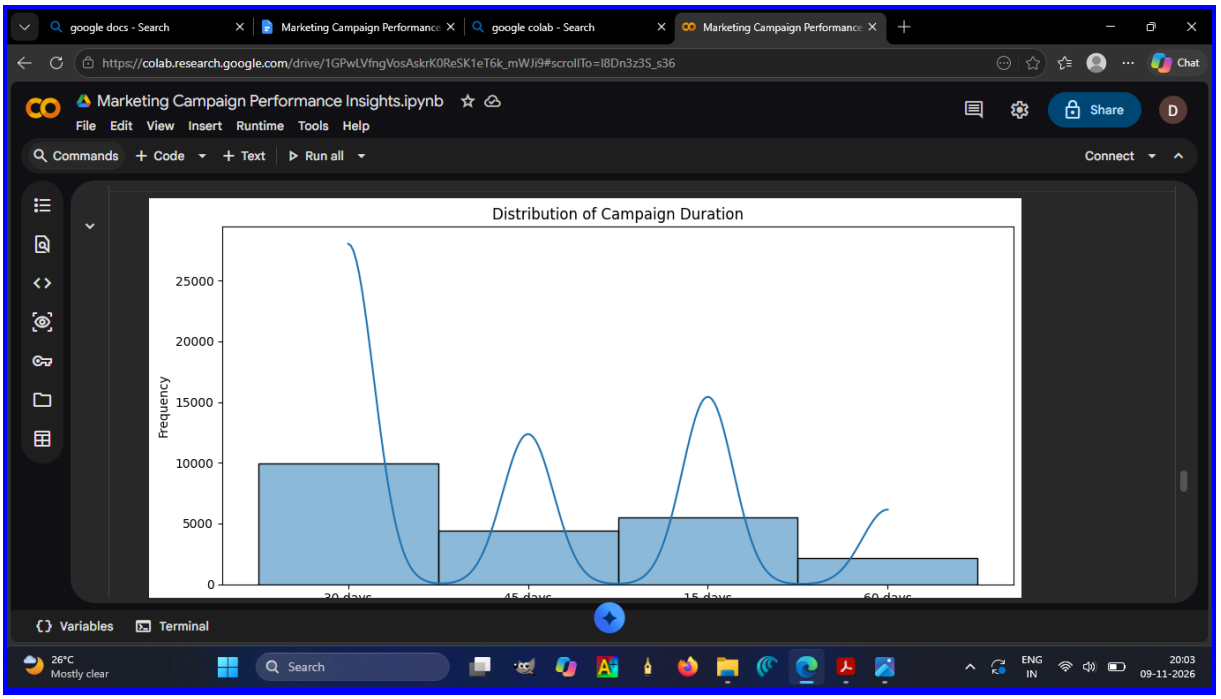
- For comparing the Engagement_Score for different Channels_Used, segmented by Campaign_Type, using a bar chart by calculating engagement by using groupby and mean methods with reset indexing methods.
- For visualizing the distribution of total ROI across different Channels_Used ,I had used a pie chart .
- For creating a scatter plot to show the relationship between Clicks and Impressions for each Channel_Used. I had assigned the clicks in x axis and impression in y axis and channel used in hue.
- The above mentioned screenshots represent the results.

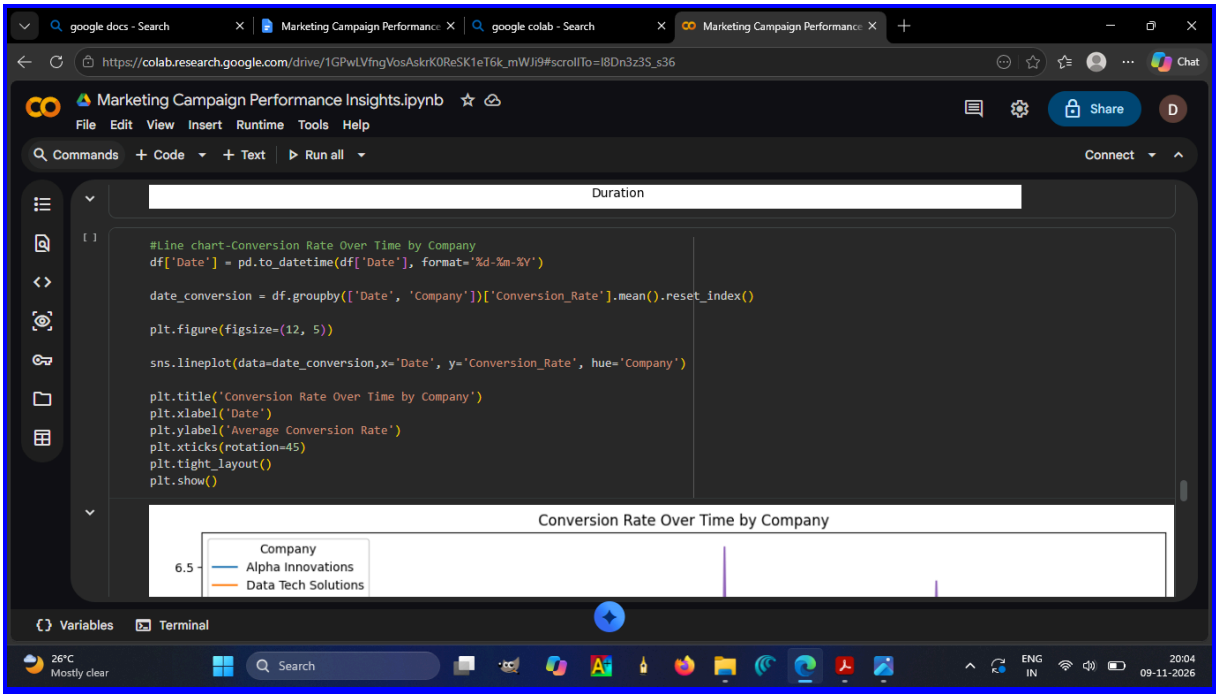
Task 4:Time-Based Analysis

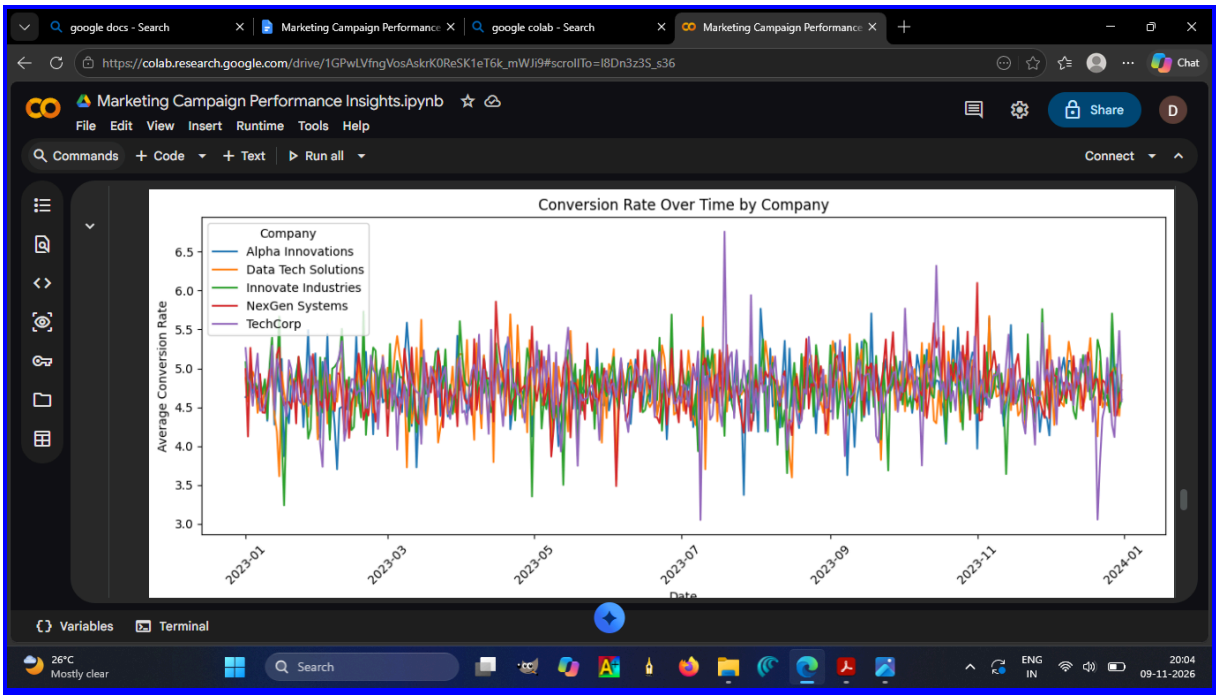
Solution :

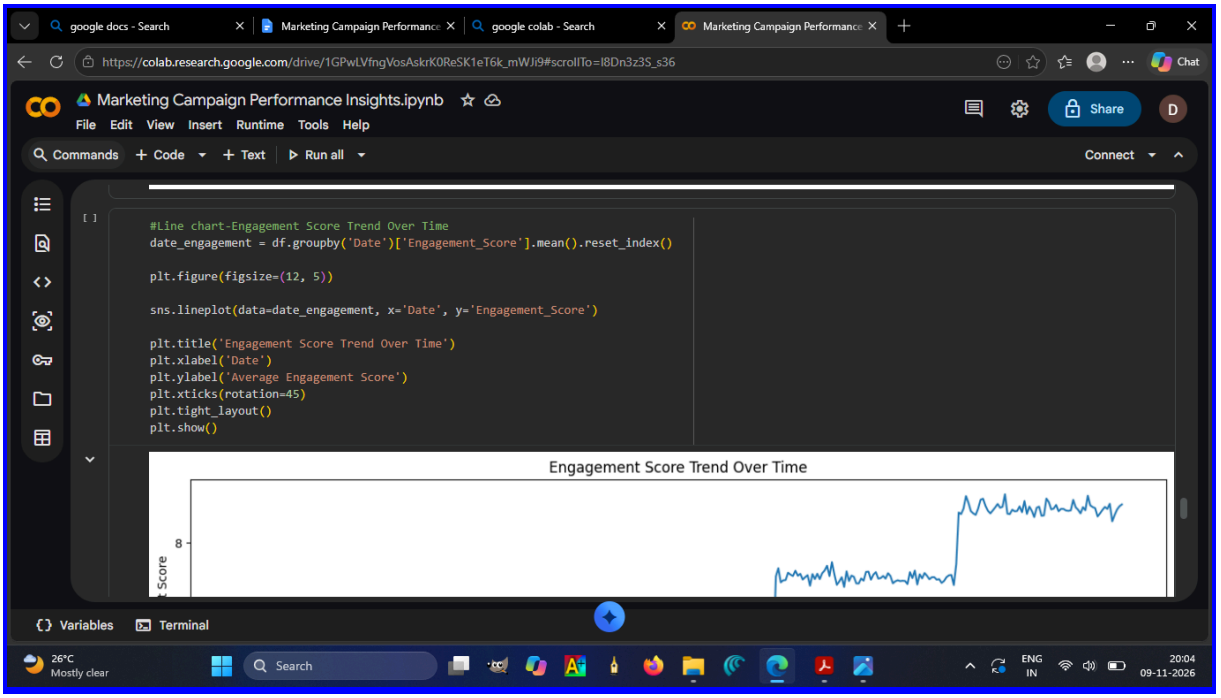
Preview screenshot :

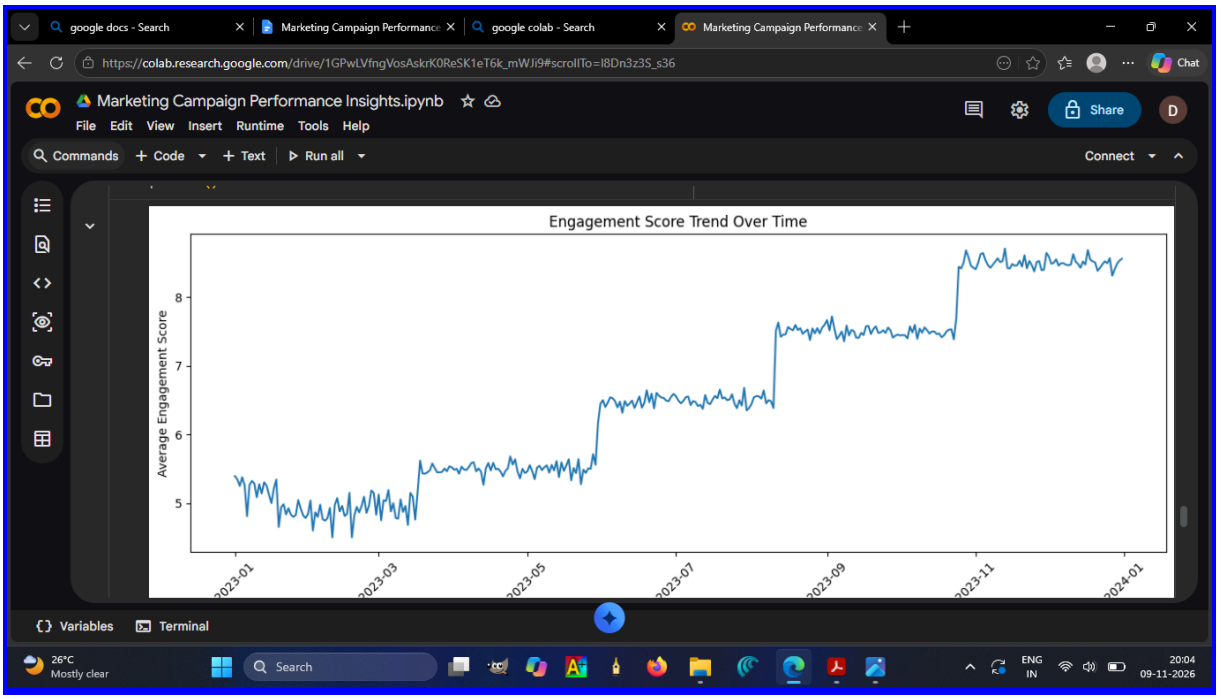





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Duration
#Line chart-Conversion Rate Over Time by Company
df['Date'] = pd.to_datetime(df['Date'], format='%d-%m-%Y')
date_conversion = df.groupby(['Date', 'Company'])['Conversion_Rate'].mean().reset_index()
plt.figure(figsize=(12, 5))
sns.lineplot(data=date_conversion,x='Date', y='Conversion_Rate', hue='Company')
plt.title('Conversion Rate Over Time by Company')
plt.xlabel('Date')
plt.ylabel('Average Conversion Rate')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
Conversion Rate Over Time by Company
Company
Alpha Innovations
Data Tech Solutions
6.5
Variables
Terminal


Marketing Campaign Performance Insights.ipynb
File Edit View Insert Runtime Tools Help
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Connect
Share
Conversion Rate Over Time by Company
Company
Alpha Innovations
Data Tech Solutions
Innovate Industries
NexGen Systems
TechCorp
Average Conversion Rate
6.5
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2023-01
2023-03
2023-05
2023-07
2023-09
2023-11
2024-01
Date
Variables
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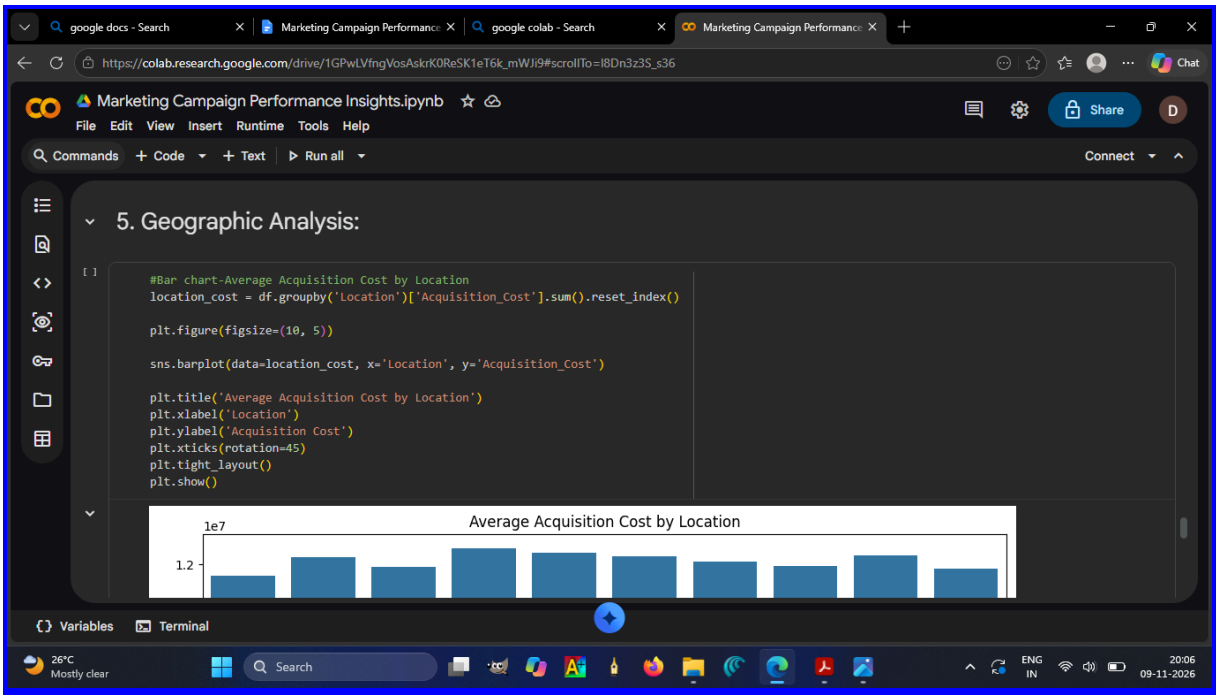
Explanation :

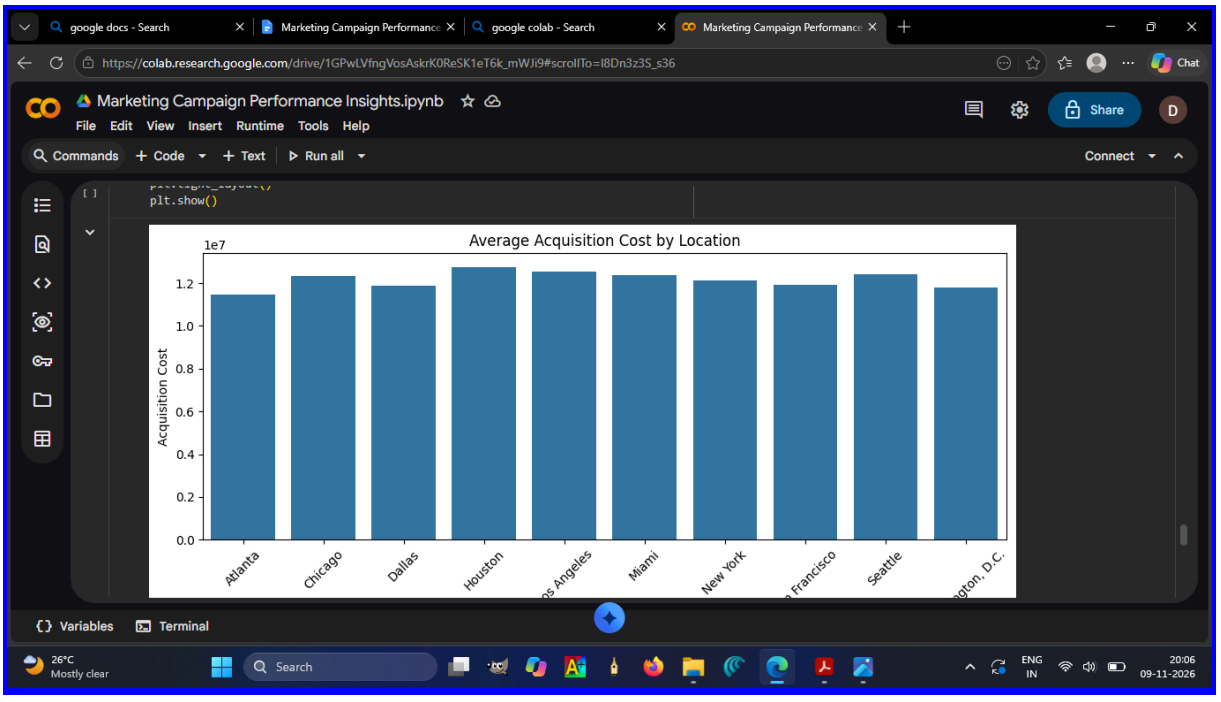
- To find the distribution of campaign duration ,I used the histogram visualization .
- To analyse how the overall Conversion_Rate has changed over Date for each Company , I had used the line chart visual.
- To examine the trend of Engagement_Score over Date, I used the line chart by assigning date in x axis and engagement score in y axis.
- The above mentioned screenshots represent the results.

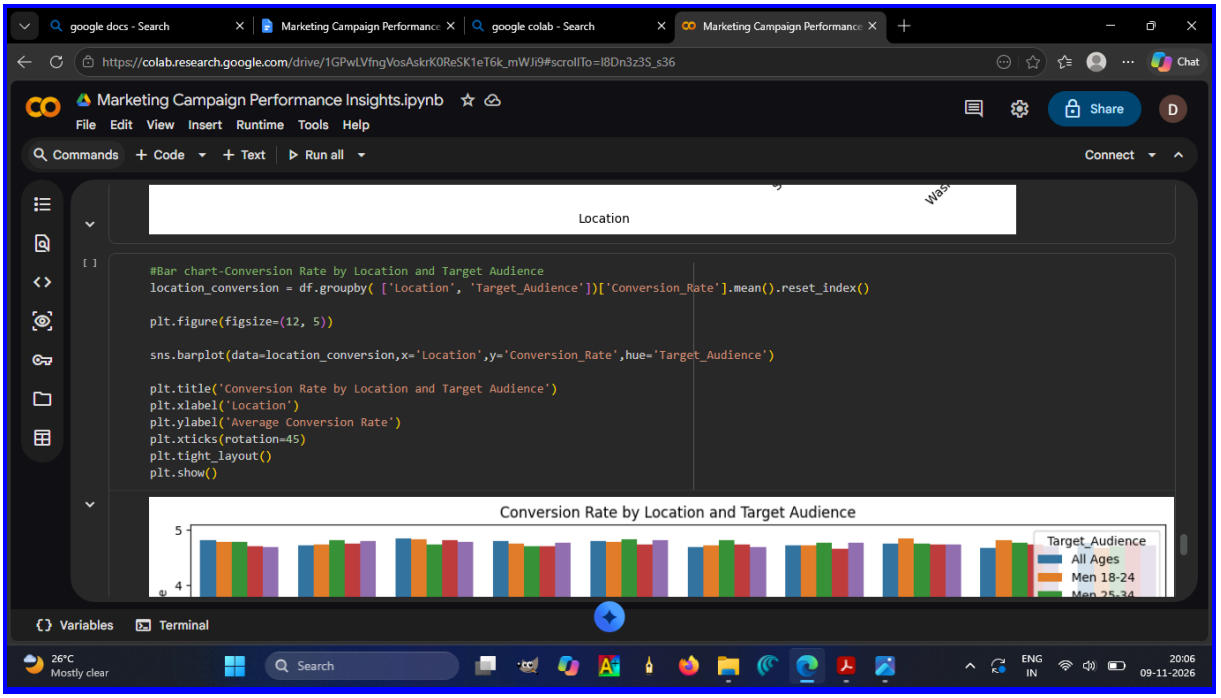
Task 5 : Geographic Analysis

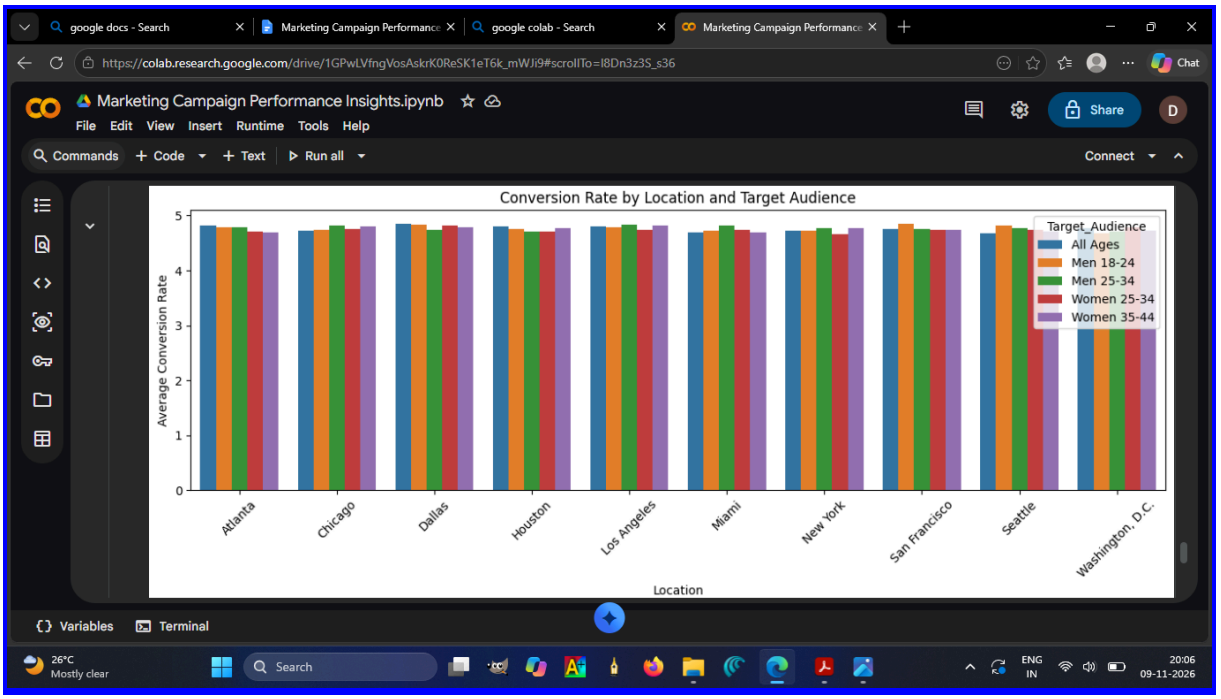
Solution :

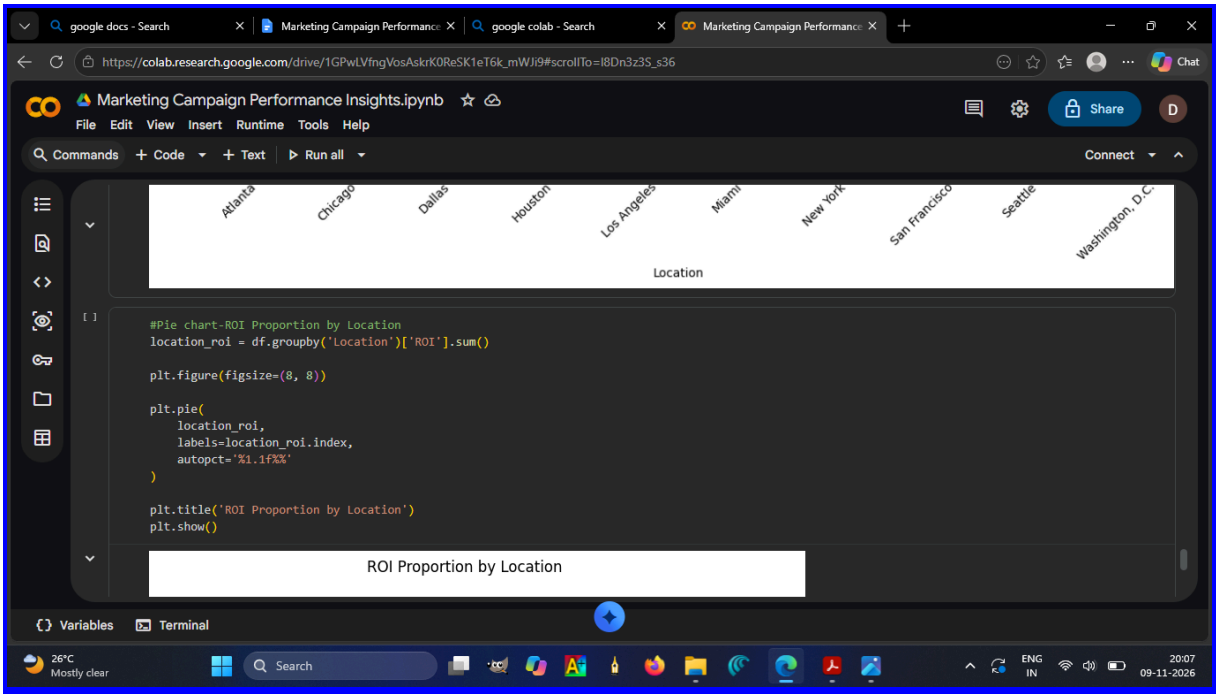
Preview screenshot :

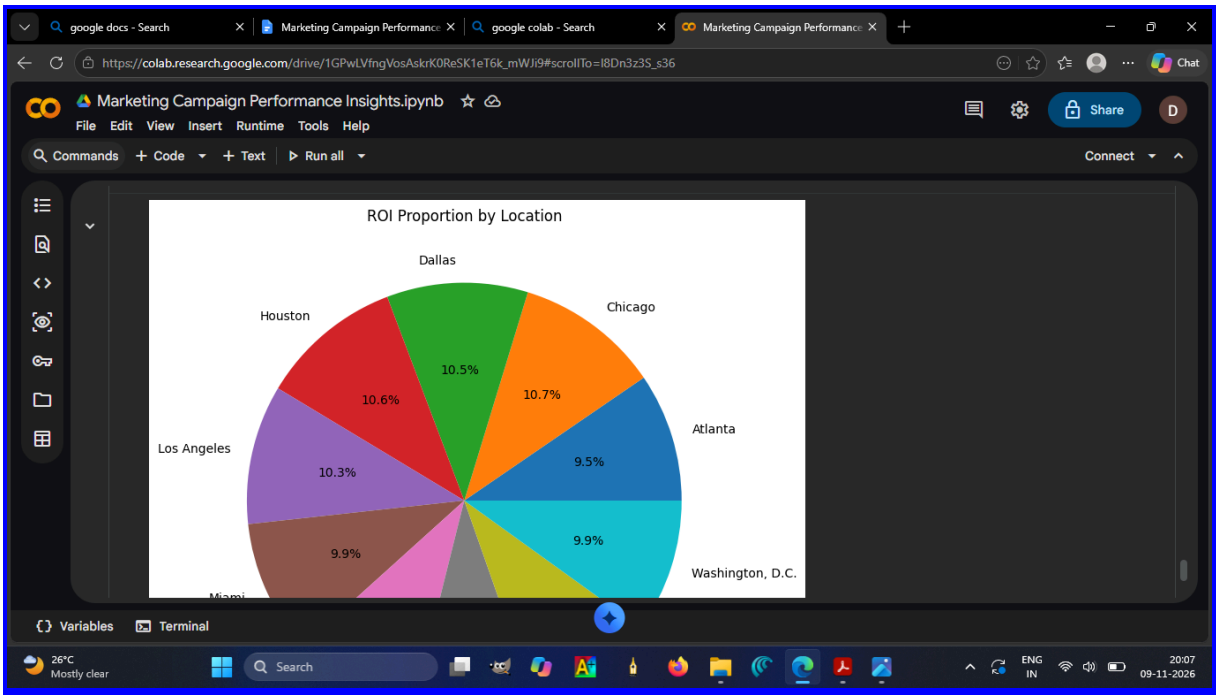





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Connect
Location
#Bar chart-Conversion Rate by Location and Target Audience
location_conversion = df.groupby(['Location', 'Target_Audience'])['Conversion_Rate'].mean().reset_index()
plt.figure(figsize=(12, 5))
sns.barplot(data=location_conversion,x='Location',y='Conversion_Rate',hue='Target_Audience')
plt.title('Conversion Rate by Location and Target Audience')
plt.xlabel('Location')
plt.ylabel('Average Conversion Rate')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
Conversion Rate by Location and Target Audience
Target_Audience
All Ages
Men 18-24
Variables
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Conversion Rate by Location and Target Audience
Target_Audience
All Ages
Men 18-24
Men 25-34
Women 25-34
Women 35-44
Average Conversion Rate
Atlanta
Chicago
Dallas
Houston
Los Angeles
Miami
New York
San Francisco
Seattle
Washington, D.C.
Location
Variables
Terminal
26°C
Mostly clear
Search
ENG
IN
20:06
09-11-2026





Explanation :

- To determine which location has the highest Acquisition_Cost . I used the bar chart and assigned location in x axis and acquisition cost in y axis.
- To visualize the Conversion_Rate by different Location, categorized by Target_Audience, I used the bar chart and assigned location in x axis , conversion rate in y axis and target audience in hue.
- To illustrate the proportion of ROI by Location. I used to create pie charts for visualization.
- The above mentioned screenshots represent the results.

Insight Summary:

- Low cost = high ROI. Campaigns costing ~1,000-3,000 give the best returns.
- Conversion is the same everywhere. Channels, companies, cities and languages all stay around 4.7-4.8 (English is a bit higher at ~5.0).
- Engagement went up in 2023, but conversion did not. They are almost unrelated (0.011).
- Best audience: Women 25-34. Best for engagement: Social Media.